# MMseqs2: The Missing Manual

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# Contents

# 1 MMseqs2 Documentation Overview

MMseqs2 is a cascaded system for large-scale sequence analysis. User-facing workflows are orchestration layers that route data through lower-level modules with strict database contracts, resource tradeoffs, and mode-specific semantics. This manual follows the same architecture so readers can move from intent to execution detail without switching mental models.

At a high level, MMseqs2 gains speed by combining three design choices: compact internal databases, staged candidate reduction before expensive alignment, and parallel execution across cores and servers. Accuracy, runtime, and resource usage are therefore not independent concerns. They are coupled through the pipeline path, indexing strategy, and split policy selected for a run.

## 1.1 Reading Path

| Stage | Goal | Entry Point |
|---|---|---|
| Overall idea | Understand how MMseqs2 turns large search spaces into tractable stages | `introduction.md` |
| Sharp bits | Avoid expensive correctness and performance mistakes | `sharp_bits.md` |
| System view | See API layers and dependency topology | `system_map.md` |
| Performance foundations | Understand storage, indexing, memory-split tradeoffs, and parallel execution model | `foundations.md` |
| Functional modules | Navigate by task domain and submodule relationships | `manual.md`, `submodules/*.md` |
| Expert behavior | Compose robust custom pipelines and maintain doc quality | `expert_manual.md` |

## 1.2 Cascade Model

A typical MMseqs2 pipeline starts with database preparation, narrows candidates through prefiltering, computes alignment or rescoring on the reduced candidate set, and then applies downstream transforms such as clustering, taxonomy assignment, or export. This staged execution is the core scaling mechanism: expensive operations are deferred until cheap filters reduce the search space.

For practical work, this means that most large performance wins come from upstream decisions, not late-stage tweaking. Index reuse, split strategy, and load behavior can dominate total wall time before sensitivity or filtering changes even become visible.

### 1.3 Navigation Axes

The documentation exposes two orthogonal views of the same commands. The functional view answers what to run for a task; the API-layer view answers where that command sits in the cascade and what it depends on.

| View | Question It Answers | Main Files |
|---|---|---|
| Functional modules | Which commands solve this workflow goal? | `manual.md`, `submodules/*.md` |
| Dependency and API layer | Which modules call this command, and which modules it calls | `system_map.md`, `reference/dependency_map.md`, `reference/index.md` |

**Note**

When narrative text and command defaults disagree, treat local help snapshots in `mmseqs_help_output` as canonical for CLI behavior. Treat `foundations.md` as the canonical summary of storage and parallel mechanics in this manual.

### 1.4 How to Use This Manual Efficiently

If you are selecting commands for a known task, start with `manual.md` and then open the linked submodule page. If you are diagnosing runtime, memory, or output interpretation problems, read `system_map.md` and `foundations.md` first, then inspect dependency links in `reference/dependency_map.md`. If you are composing custom pipelines, finish with `expert_manual.md` for contract and reproducibility discipline.

### 1.5 Source of Truth

| Topic | Canonical Source |
|---|---|
| Visible command set and one-line intent | `MMseqs2/src/MMseqsBase.cpp` |
| Workflow orchestration logic | `MMseqs2/src/workflow/*.cpp`, `MMseqs2/data/workflow/*.sh` |
| Exact options and defaults | `mmseqs_help_output/*.txt` |
| Crosslinked command/module references | `mmseqs2_docs/reference/`, `mmseqs2_docs/submodules/` |
| Legacy long-form historical material | `mmseqs2_docs/wiki.md` |

## 2 Sharp Bits

Most expensive MMseqs2 failures are not algorithmic bugs. They come from contract mismatches, mode drift across runs, and late detection of pipeline issues. The goal of this section is to front-load the mistakes that cost the most time in large-scale runs.

The recurring pattern is simple: workflows appear similar, but runtime and output behavior are shaped by earlier infrastructure choices such as indexing, load mode, split policy, and sidecar completeness. A small mismatch upstream can invalidate interpretation downstream.

| Risk Surface | Why It Matters | Guardrail |
|---|---|---|
| Rebuilding indexes every run | Repeated setup work dominates runtime on stable targets | Reuse `createindex` or `createlinindex` artifacts |
| Raising sensitivity too early | Runtime and intermediate volume increase quickly | Tune indexing and split policy before raising `-s` |
| Split-memory-temp mismatch | Reduced RAM can increase I/O and wall time | Tune `--split*` with both memory and temporary disk budgets |
| Mixed alignment/rescore modes | Output fields become semantically incomparable | Lock mode flags when comparing experiments |
| DB sidecar mismatches | Downstream modules silently lose required metadata | Validate `_h`, taxonomy, and lookup sidecars before chaining |
| Taxonomy and ORF preprocessing drift | Classification quality changes with preprocessing choices | Fix frame, ORF, and aggregation policy per dataset type |
| Duplicate conversion steps | Unnecessary transforms add cost and interpretation noise | Keep a minimal transform chain with explicit rationale |

## 2.1 Diagnostics Sequence

When something looks wrong, inspect in this order: database contracts, mode consistency, resource policy, then threshold tuning. This order prevents expensive false debugging loops where parameter changes mask an infrastructure mismatch.

**Performance**

Treat performance tuning as a sequence: index reuse first, split-memory policy second, sensitivity increase last.

**Warning**

Treat exported tables as pipeline artifacts, not standalone truth. Interpretation depends on upstream search mode, alignment mode, and filters.

**Tip**

Validate pipelines on a small representative subset before scaling to full datasets. Early contract checks prevent full reruns.

# 3 System Map: Cascaded APIs and Module Connections

MMseqs2 commands are layered. Workflow and high-level commands orchestrate lower-level modules, and those modules enforce the actual data contracts. This layering is both an architecture map and a performance map: most runtime and correctness characteristics are inherited from the lower levels each workflow chooses.

Primary generated maps are `reference/index.md` and `reference/dependency_map.md`.

## 3.1 API Layers

| Layer | Typical Role | Representative Commands |
|---|---|---|
| `workflow` | Easy entry points over FASTA/FASTQ with default orchestration | `easy-search`, `easy-cluster`, `easy-taxonomy` |
| `high_level_api` | End-to-end MMseqs DB workflows for search, clustering, taxonomy, or set workflows | `search`, `cluster`, `taxonomy`, `linsearch`, `multihitsearch` |
| `mid_level_api` | Core compute modules used inside workflows | `prefilter`, `align`, `clust`, `kmermatcher` |
| `low_level_api` | DB operations, conversion, utilities, and composition helpers | `createdb`, `convertalis`, `filterdb`, `createtsv` |

## 3.2 Cascade Pattern

Most pipelines follow the same execution shape: an orchestration command selects a path, prefiltering narrows candidate pairs, alignment or rescoring computes pair quality, and downstream modules transform or aggregate outputs. Not every command uses every stage, but this pattern explains most runtime and output behavior.

The important operational consequence is that changing a high-level workflow can silently switch lower-level modules and therefore change both output semantics and resource profile. Dependency links should be read as behavior links, not only call graph links.

**Note**

Functional grouping and API layer are orthogonal. A command can be in the taxonomy group and still be low-level API.

### 3.3 Crosslink Model Used in Submodule Pages

Each command entry in `submodules/*.md` includes a structured metadata table with:

| Field | Purpose |
|---|---|
| API layer and category | Places command in the execution stack |
| Called-by and calls | Exposes upstream and downstream dependencies |
| Related functional groups | Shows cross-domain coupling |
| Workflow script usage | Shows script-level evidence from extracted workflows |
| Reference links | Links to full CLI help and dependency-map anchor |

### 3.4 When to Start from the Dependency View

Start with `reference/dependency_map.md` when tuning runtime at workflow level, debugging unexpected output semantics, or composing custom pipelines from low-level modules. Start with `manual.md` and submodule pages when task intent is clear and you need command selection guidance first.

### 3.5 Transition to Performance Foundations

This chapter explains where commands sit in the cascade. The next chapter, `foundations.md`, explains why that cascade is fast in practice: internal storage format, index and load mechanics, memory and split tradeoffs, and parallel execution strategies.

## 4 Performance Foundations: Why MMseqs2 Is Fast

This chapter summarizes the low-level mechanics that make MMseqs2 fast at scale. It is the canonical overview for storage format, indexing behavior, memory-split tradeoffs, and parallel execution strategy in this manual.

### 4.1 1. Internal Database and Storage Model

Most MMseqs2 modules consume and produce MMseqs2 database files rather than plain FASTA. The design avoids creating millions of small files and instead stores records in contiguous files with explicit indexing.

| Component | Role | Performance Implication |
|---|---|---|
| `<db>` data file | Concatenated records separated by `\0` | Sequential storage keeps filesystem overhead low |
| `<db>.index` | ID, byte offset, and record size per entry | Enables direct random access by key |

| Component | Role | Performance Implication |
|---|---|---|
| `<db>.dbtype` | Binary type marker (protein, nucleotide, profile, generic, etc.) | Enforces module input contracts early |
| `<db>_h` and `<db>_h.index` | Header sidecar database | Separates sequence-heavy and header-heavy access paths |
| `<db>.lookup` and taxonomy sidecars | External ID and annotation mapping | Required for complete downstream annotation workflows |

Because offsets in `.index` point into immutable byte positions, in-place editing is avoided. MMseqs2 generally writes new databases for transformations. This design preserves indexing correctness and enables predictable high-throughput I/O.

> **Note**
>
> Header and sequence separation is a deliberate performance choice. Sequence-heavy stages can avoid header I/O, while export and annotation stages can load both paths explicitly.

## 4.2 2. Indexing and Data-Access Strategy

Precomputed indexes improve repeated searches against stable targets by shifting cost from repeated startup to reusable artifacts. The gain is strongest when targets are reused many times or iterative workflows are run.

The placement of index data matters. If index files are accessed through slower shared storage, read-in can become a bottleneck. In those cases, a strategy that keeps index data resident in memory, or in local fast storage, can outperform repeated shared-storage reads.

MMseqs2 load behavior also matters:

| Access Pattern | Typical Effect |
|---|---|
| Precomputed index with efficient local access | Fast startup across repeated runs |
| Shared-storage index read bottleneck | Startup overhead dominates short queries |
| Memory-resident index path | Reduces repeated transfer overhead |
| Alternative load modes (`--db-load-mode`) | Trade off startup copy behavior and runtime access pattern |

For small query sets, memory-mapped access can minimize startup latency if target index data is already resident. For large query sets, alternative copy behavior can improve throughput by reducing translation-lookaside-buffer pressure in prefilter-heavy phases.

## 4.3 3. Memory Model and Split Tradeoffs

Prefiltering is usually the largest memory consumer. Legacy MMseqs2 guidance models prefilter index memory approximately as:

```
M = (7 * N * L + 8 * a^k) bytes
```

where `N` is sequence count, `L` is average sequence length, `a` is alphabet size, and `k` is k-mer length. The key operational takeaway is linear growth with database size plus an exponential alphabet-and-k-mer term for pointer structures.

When memory is insufficient, MMseqs2 splits databases and processes chunks. Splitting lowers peak memory but introduces additional merge and I/O overhead.

| Split Choice | Benefit | Cost |
|---|---|---|
| Fewer/larger chunks | Better runtime efficiency | Higher peak memory requirement |
| More/smaller chunks | Lower peak memory footprint | More merge and temporary I/O overhead |
| `--split-memory-limit` cap | Predictable memory envelope | Usually slower wall time than unsplit runs |

The slowdown from splitting is often less visible at high sensitivity, where core compute dominates, and more visible at low sensitivity, where merge and I/O overhead are a larger share of total runtime.

## 4.4 4. Parallel Execution Model

MMseqs2 parallelism is layered:

| Layer | Mechanism | Main Idea |
|---|---|---|
| Intra-node | Multi-core execution (OpenMP-based stages) | Parallelize computational kernels within each process |
| Inter-node | MPI runner model | Distribute chunks across servers, then execute local multi-core work |
| Scheduler-based | Batch-system splitting | Partition query DB and submit independent jobs when MPI is not used |

In MPI mode, workloads can be distributed by splitting targets or queries. Target splitting usually reduces per-node memory pressure but can be less time-efficient because merging is I/O-heavy. Query splitting can be easier to distribute in batch environments but may duplicate some target-side work.

Temporary storage placement is also part of parallel performance. Shared temporary storage simplifies coordination but can create contention at high split counts. Local temporary storage can reduce shared-disk bottlenecks in large distributed runs.

> **Warning**
>
> Distributed performance is constrained by I/O topology as much as by CPU count. Evaluate shared-disk behavior before attributing slowdowns to compute.

### 4.5 5. Practical Tuning Order

A stable tuning sequence avoids expensive trial-and-error:

| Step | Question |
|---|---|
| 1. Contract check | Are DB types and sidecars compatible across the pipeline? |
| 2. Index strategy | Are stable targets indexed and loaded in a way that matches storage topology? |
| 3. Split and temp policy | Is memory capped appropriately without overpaying in merge and I/O overhead? |
| 4. Parallel model | Is the run mode aligned with hardware and storage architecture? |
| 5. Sensitivity and filters | Only after infrastructure choices are stable, are scoring thresholds tuned |

This ordering reflects how MMseqs2 runtime is usually determined in practice: infrastructure first, algorithmic strictness second.

### 4.6 6. Cross References

Use `manual.md` and `submodules/*.md` for task-oriented command selection, `reference/dependency_map.md` for call-topology debugging, and `expert_manual.md` for advanced composition and reproducibility discipline. Use `wiki.md` for historical long-form detail and extended examples beyond this condensed chapter.

## 5 Functional Modules Manual

This section is the task-oriented navigation hub. Each functional page groups related commands, then links every command to API layer and dependency context so you can move directly to execution details.

Before diving into module pages for large production runs, read `foundations.md`. That chapter explains storage, indexing, split policy, and parallel model assumptions that shape runtime and output behavior across all modules.

## 5.1 Functional Groups

| Functional Group | Scope | Module Page |
|---|---|---|
| Easy Workflows | Shortcut workflows over FASTA/FASTQ with standard outputs | Easy Workflows |
| Search Workflows | Homology search and mapping workflows | Search Workflows |
| Clustering | Cascaded and linear clustering workflows and cluster transforms | Clustering |
| Prefiltering | Candidate-generation modules before expensive alignment | Prefiltering |
| Alignment | Alignment and rescoring modules | Alignment |
| Profiles | Profile and MSA creation, conversion, and profile-search helpers | Profiles |
| Database Management | DB creation, indexing, merging, splitting, and storage operations | Database Management |
| Result Handling | Result filtering, summarization, and export transforms | Result Handling |
| Sequence Manipulation | ORF, frame, masking, and sequence-level transforms | Sequence Manipulation |
| Taxonomy | Taxonomy DB preparation, assignment, filtering, and reports | Taxonomy |
| Multi-hit | Set-based search and aggregation workflows | Multi-hit |
| Utilities | General and special-purpose helpers used in custom pipelines | Utilities |

## 5.2 Complementary Maps

Functional pages answer what to run. Generated maps answer how commands connect.

| Need | Open |
|---|---|
| Layered architecture and navigation rules | System Map |
| Storage, indexing, memory-split, and parallel execution model | Performance Foundations |
| Complete command catalog with CLI pages | Command Reference Index |
| Command dependency topology | Dependency Map |

# 6 Easy Workflows

High-level shortcuts that operate directly on FASTA/FASTQ and produce user-facing outputs with minimal setup.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

> **Performance**
>
> For repeated runs against stable targets, prioritize index reuse and split-memory tuning before increasing sensitivity.

## 6.1 `easy-cluster`

Slower, sensitive clustering.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs easy-cluster <i:fastaFile1[.gz\|.bz2]> ... <i:fastaFileN[.gz\|.bz2]> <o:clusterPrefix> <tmpDir> [options]` |
| API layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Called by modules | `n/a` |
| Calls modules | `cluster`, `createdb`, `createseqfiledb`, `createtsv`, `result2flat`, `result2repseq`, `rmdb` |
| Related functional groups | `clustering`, `database`, `result_handling` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 6.1.1 Key Options

| Option | Purpose |
|---|---|
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |
| `--alph-size` | Alphabet size (range 2-21) |
| `--max-seqs` | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| `--split` | Split input into N equally distributed chunks. 0: set the best split automatically |

## 6.2 `easy-linclust`

Fast linear time cluster, less sensitive clustering.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs easy-linclust <i:fastaFile1[.gz\|.bz2]> ... <i:fastaFileN[.gz\|.bz2]> <o:clusterPrefix> <tmpDir> [options]` |
| API layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Called by modules | `n/a` |
| Calls modules | `createdb`, `createseqfiledb`, `createtsv`, `linclust`, `result2flat`, `result2repseq`, `rmdb` |
| Related functional groups | `clustering`, `database`, `result_handling` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 6.2.1 Key Options

| Option | Purpose |
|---|---|
| --comp-bias-corr | Correct for locally biased amino acid composition (range 0-1) |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |
| --add-self-matches | Artificially add entries of queries with themselves (for clustering) |
| --alph-size | Alphabet size (range 2-21) |
| --spaced-kmer-mode | 0: use consecutive positions in k-mers; 1: use spaced k-mers |
| --spaced-kmer-pattern | User-specified spaced k-mer pattern |
| --mask | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob | Mask sequences is probablity is above threshold |

## 6.3 easy-linsearch

Fast, less sensitive homology search.

| Aspect | Value |
|---|---|
| Usage | usage: mmseqs easy-linsearch <i:queryFastaFile1[.gz\|.bz2]> ... <i:queryFastaFileN[.gz\|.bz2]> <i:targetFastaFile[.gz\|.bz2]>\| <i:targetDB> <o:alignmentFile> <tmpDir> [options] |
| API layer | workflow |
| Category flags | COMMAND_EASY \| COMMAND_EXPERT |
| Called by modules | n/a |
| Calls modules | convertalis, createdb, createlinindex, linsearch, rmdb, search, summarizeresult |
| Related functional groups | database, result_handling, search_workflows |
| Workflow script usage | n/a |

Reference links: Full CLI, Dependency map.

### 6.3.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `--mask` | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| `--mask-prob` | Mask sequences is probablity is above threshold |
| `--mask-lower-case` | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region |
| `--mask-n-repeat` | Repeat letters that occure > threshold in a rwo |

## 6.4 `easy-rbh`

Find reciprocal best hit.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs easy-rbh <i:queryFastaFile1[.gz\|.bz2]> <i:targetFastaFile[.gz\|.bz2]>\| <i:targetDB> <o:alignmentFile> <tmpDir> [options]` |
| API layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Called by modules | `n/a` |
| Calls modules | `convertalis`, `createdb`, `rbh`, `rmdb` |
| Related functional groups | `database`, `result_handling`, `search_workflows` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 6.4.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |

## 6.5 `easy-search`

Sensitive homology search.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs easy-search <i:queryFastaFile1[.gz\|.bz2]> ... <i:queryFastaFileN[.gz\|.bz2]>\| <i:stdin> <i:targetFastaFile[.gz]>\| <i:targetDB> <o:alignmentFile> <tmpDir> [options]` |
| API layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Called by modules | `n/a` |
| Calls modules | `convertalis`, `createdb`, `createlinindex`, `linsearch`, `rmdb`, `search`, `summarizeresult` |
| Related functional groups | `database`, `result_handling`, `search_workflows` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 6.5.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |

## 6.6 `easy-taxonomy`

Taxonomic classification.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs easy-taxonomy <i:fastaFile1[.gz\|.bz2]> ... <i:fastaFileN[.gz\|.bz2]> <i:targetDB> <o:taxReports> <tmpDir> [options]` |
| API layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Called by modules | `n/a` |
| Calls modules | `addtaxonomy`, `convertalis`, `createdb`, `createtsv`, `filterdb`, `lca`, `rmdb`, `summarizealis`, `swapresults`, `taxonomy`, `taxonomyreport` |
| Related functional groups | `database`, `result_handling`, `taxonomy`, `utilities` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 6.6.1 Key Options

| Option | Purpose |
| --- | --- |
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |

# 7 Search Workflows

Workflow-level search and mapping modules that orchestrate prefiltering and alignment under different modes.

**Note**

This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

**Performance**

For repeated runs against stable targets, prioritize index reuse and split-memory tuning before increasing sensitivity.

## 7.1 `linsearch`

Fast, less sensitive homology search.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs linsearch <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MAIN\|COMMAND_EXPERT` |
| Called by modules | `easy-linsearch`, `easy-search` |
| Calls modules | `align`, `concatdbs`, `extractorfs`, `filterdb`, `kmersearch`, `offsetalignment`, `rescorediagonal`, `rmdb`, `swapresults` |
| Related functional groups | `alignment`, `database`, `easy_workflows`, `prefiltering`, `result_handling`, `sequence_manipulation`, `utilities` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 7.1.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `--mask` | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| `--mask-prob` | Mask sequences is probablity is above threshold |
| `--mask-lower-case` | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region |
| `--mask-n-repeat` | Repeat letters that occure > threshold in a rwo |

## 7.2 `map`

Map nearly identical sequences.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs map <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Called by modules | `n/a` |
| Calls modules | `search` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 7.2.1 Key Options

| Option | Purpose |
|---|---|
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |
| `--alph-size` | Alphabet size (range 2-21) |
| `--max-seqs` | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| `--split` | Split input into N equally distributed chunks. 0: set the best split automatically |

## 7.3 `rbh`

Reciprocal best hit search.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs rbh <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Called by modules | `easy-rbh` |
| Calls modules | `filterdb`, `mergedbs`, `result2rbh`, `rmdb`, `search`, `swapresults` |
| Related functional groups | `database`, `easy_workflows`, `result_handling`, `utilities` |
| Workflow script usage | `easyrbh.sh` |

Reference links: Full CLI, Dependency map.

### 7.3.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |

## 7.4 `search`

Sensitive homology search.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs search <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Called by modules | `clusterupdate`, `easy-linsearch`, `easy-search`, `map`, `multihitsearch`, `rbh`, `search`, `taxonomy` |
| Calls modules | `align`, `createsubdb`, `expand2profile`, `expandaln`, `extractframes`, `extractorfs`, `filterresult`, `lcaalign`, `mergedbs`, `mvdb`, `offsetalignment`, `prefilter`, `profile2consensus`, `rescorediagonal`, `result2profile`, `result2stats`, `rmdb`, `search`, `splitsequence`, `subtractdbs`, `swapresults`, `ungappedprefilter` |
| Related functional groups | `alignment`, `clustering`, `database`, `easy_workflows`, `multi_hit`, `prefiltering`, `profiles`, `result_handling`, `sequence_manipulation`, `taxonomy` |
| Workflow script usage | `enrich.sh`, `iterativepp.sh`, `map.sh`, `multihitsearch.sh`, `rbh.sh`, `taxonomy.sh`, `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 7.4.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |

| Option | Purpose |
|---|---|
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |

# 8 Clustering

Modules for cluster construction, updates, and representative handling across different clustering strategies.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

> **Performance**
>
> For repeated runs against stable targets, prioritize index reuse and split-memory tuning before increasing sensitivity.

## 8.1 `clust`

Cluster result by Set-Cover/Connected-Component/Greedy-Incremental.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs clust <i:sequenceDB> <i:resultDB> <o:clusterDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_CLUSTER` |
| Called by modules | `cluster`, `linclust` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `cascaded_clustering.sh`, `clustering.sh`, `linclust.sh`, `nucleotide_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 8.1.1 Key Options

| Option | Purpose |
| --- | --- |
| `--cluster-mode` | 0: Set-Cover (greedy) |
| `--max-iterations` | Maximum depth of breadth first search in connected component clustering |
| `--similarity-type` | Type of score used for clustering. 1: alignment score 2: sequence identity |
| `--weights` | Weights used for cluster priorization |
| `--cluster-weight-threshold` | Weight threshold used for cluster priorization |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 8.2 `cluster`

Slower, sensitive clustering.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs cluster <i:sequenceDB> <o:clusterDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Called by modules | `clusterupdate`, `easy-cluster` |
| Calls modules | `align`, `clust`, `clusthash`, `concatdbs`, `createsubdb`, `extractframes`, `filterdb`, `linclust`, `mergeclusters`, `mergedbs`, `mvdb`, `offsetalignment`, `prefilter`, `rescorediagonal`, `rmdb`, `subtractdbs`, `swapdb`, `tsv2db` |
| Related functional groups | `alignment`, `database`, `easy_workflows`, `prefiltering`, `sequence_manipulation`, `utilities` |
| Workflow script usage | `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 8.2.1 Key Options

| Option | Purpose |
|---|---|
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |
| `--alph-size` | Alphabet size (range 2-21) |
| `--max-seqs` | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| `--split` | Split input into N equally distributed chunks. 0: set the best split automatically |

## 8.3 `clusterupdate`

Update previous clustering with new sequences.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs clusterupdate <i:oldSequenceDB> <i:newSequenceDB> <i:oldClustResultDB> <o:newMappedSequenceDB> <o:newClustResultDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Called by modules | `n/a` |
| Calls modules | `cluster`, `concatdbs`, `createsubdb`, `diffseqdbs`, `filterdb`, `mergedbs`, `mvdb`, `prefixid`, `renamedbkeys`, `result2repseq`, `rmdb`, `search`, `swapdb` |
| Related functional groups | `database`, `result_handling`, `search_workflows`, `utilities` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 8.3.1 Key Options

| Option | Purpose |
| --- | --- |
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |

## 8.4 `clusthash`

Hash-based clustering of equal length sequences.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs clusthash <i:sequenceDB> <o:alignmentDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_CLUSTER` |
| Called by modules | `cluster` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `clustering.sh` |

Reference links: Full CLI, Dependency map.

### 8.4.1 Key Options

| Option | Purpose |
| --- | --- |
| `--alph-size` | Alphabet size (range 2-21) |

| Option | Purpose |
|---|---|
| `--min-seq-id` | List matches above this sequence identity (for clustering) (range 0.0-1.0) |
| `--sub-mat` | Substitution matrix file |
| `--max-seq-len` | Maximum sequence length |
| `--db-load-mode` | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 8.5 `linclust`

Fast, less sensitive clustering.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs linclust <i:sequenceDB> <o:clusterDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Called by modules | `cluster`, `easy-linclust` |
| Calls modules | `align`, `clust`, `createsubdb`, `filterdb`, `kmermatcher`, `mergeclusters`, `rescorediagonal`, `rmdb` |
| Related functional groups | `alignment`, `database`, `easy_workflows`, `prefiltering`, `utilities` |
| Workflow script usage | `cascaded_clustering.sh`, `nucleotide_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 8.5.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |

| Option | Purpose |
|---|---|
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--alph-size` | Alphabet size (range 2-21) |
| `--spaced-kmer-mode` | 0: use consecutive positions in k-mers; 1: use spaced k-mers |
| `--spaced-kmer-pattern` | User-specified spaced k-mer pattern |
| `--mask` | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| `--mask-prob` | Mask sequences is probablity is above threshold |

### 8.6 `mergeclusters`

Merge multiple cascaded clustering steps.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs mergeclusters <i:sequenceDB> <o:clusterDB> <i:clusterDB1> ... <i:clusterDBn> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_CLUSTER` |
| Called by modules | `cluster`, `linclust` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `cascaded_clustering.sh`, `clustering.sh`, `linclust.sh`, `nucleotide_clustering.sh` |

Reference links: Full CLI, Dependency map.

#### 8.6.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 8.7 `pickconsensusrep`

Select new representatives for each cluster based on consensus.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_CLUSTER` |
| Called by modules | `n/a` |
| Calls modules | `align`, `msa2profile`, `prefixid`, `renamedbkeys`, `result2msa`, `rmdb`, `tsv2db` |
| Related functional groups | `alignment`, `database`, `profiles`, `result_handling`, `utilities` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 9 Prefiltering

Core candidate-generation modules used to reduce search space before expensive alignment stages.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

### 9.1 `countkmer`

Count k-mers.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 9.2 gappedprefilter

Optimal Smith-Waterman-based prefiltering (slow).

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs gappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 9.2.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--min-ungapped-score` | Accept only matches with ungapped alignment score above threshold |
| `--max-seqs` | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| `--gap-open` | Gap open cost |
| `--gap-extend` | Gap extension cost |
| `-e` | List matches below this E-value (range 0.0-inf) |
| `-c` | List matches above this fraction of aligned (covered) residues (see –cov-mode) |

## 9.3 kmermatcher

Find bottom-m-hashed k-mer matches within sequence DB.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs kmermatcher <i:sequenceDB> <o:prefilterDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Called by modules | `linclust` |
| Calls modules | `n/a` |
| Related functional groups | `clustering` |
| Workflow script usage | `linclust.sh` |

Reference links: Full CLI, Dependency map.

### 9.3.1 Key Options

| Option | Purpose |
| --- | --- |
| `--alph-size` | Alphabet size (range 2-21) |
| `--spaced-kmer-mode` | 0: use consecutive positions in k-mers; 1: use spaced k-mers |
| `--spaced-kmer-pattern` | User-specified spaced k-mer pattern |
| `--mask` | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| `--mask-prob` | Mask sequences is probablity is above threshold |
| `--mask-lower-case` | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region |
| `--mask-n-repeat` | Repeat letters that occure > threshold in a rwo |
| `-k` | k-mer length (0: automatically set to optimum) |

## 9.4 `kmersearch`

Find bottom-m-hashed k-mer matches between target and query DB.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs kmersearch <i:queryDB> <i:kmerIndexDB> <o:prefilterDB> [options]` |

| Aspect | Value |
|---|---|
| API layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Called by modules | `linsearch` |
| Calls modules | `n/a` |
| Related functional groups | `search_workflows` |
| Workflow script usage | `linsearch.sh` |

Reference links: Full CLI, Dependency map.

### 9.4.1 Key Options

| Option | Purpose |
|---|---|
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `--mask` | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| `--mask-prob` | Mask sequences is probablity is above threshold |
| `--mask-lower-case` | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region |
| `--mask-n-repeat` | Repeat letters that occure > threshold in a rwo |
| `--split-memory-limit` | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| `--cov-mode` | 0: coverage of query and target |
| `-c` | List matches above this fraction of aligned (covered) residues (see –cov-mode) |

## 9.5 `prefilter`

Double consecutive diagonal k-mer search.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs prefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Called by modules | `cluster`, `search`, `taxonomy` |

| Aspect | Value |
|---|---|
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `search_workflows`, `taxonomy` |
| Workflow script usage | `blastp.sh`, `blastpgp.sh`, `cascaded_clustering.sh`, `clustering.sh`, `enrich.sh`, `iterativepp.sh`, `nucleotide_clustering.sh`, `searchslicedtargetprofile.sh`, `searchtargetprofile.sh`, `taxpercontig.sh`, `translated_search.sh` |

Reference links: Full CLI, Dependency map.

### 9.5.1 Key Options

| Option | Purpose |
|---|---|
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |
| `--alph-size` | Alphabet size (range 2-21) |
| `--max-seqs` | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| `--split` | Split input into N equally distributed chunks. 0: set the best split automatically |

## 9.6 `ungappedprefilter`

Optimal diagonal score search.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs ungappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]` |
| API layer | `mid_level_api` |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_PREFILTER` |
| Called by modules | `search` |
| Calls modules | `n/a` |
| Related functional groups | `search_workflows` |
| Workflow script usage | `blastp.sh`, `blastpgp.sh` |

Reference links: Full CLI, Dependency map.

### 9.6.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--min-ungapped-score` | Accept only matches with ungapped alignment score above threshold |
| `--max-seqs` | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| `-c` | List matches above this fraction of aligned (covered) residues (see –cov-mode) |
| `-e` | List matches below this E-value (range 0.0-inf) |
| `--cov-mode` | 0: coverage of query and target |
| `--taxon-list` | Taxonomy ID, possibly multiple values separated by ',' |

# 10 Alignment

Core alignment and alignment-adjacent modules for scoring, rescoring, and coordinate transformations.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

## 10.1 `align`

Optimal gapped local alignment.

| Aspect | Value |
|---|---|
| Usage | `usage:` `mmseqs align <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Called by modules | `cluster`, `linclust`, `linsearch`, `pickconsensusrep`, `search` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `search_workflows` |
| Workflow script usage | `iterativepp.sh`, `nucleotide_clustering.sh`, `pickconsensusrep.sh`, `searchslicedtargetprofile.sh` |

Reference links: Full CLI, Dependency map.

### 10.1.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `-a` | Add backtrace string (convert to alignments with mmseqs convertalis module) |
| `--alignment-mode` | How to compute the alignment: |
| `--alignment-output-mode` | How to compute the alignment: |
| `--wrapped-scoring` | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| `-e` | List matches below this E-value (range 0.0-inf) |

## 10.2 `alignall`

Within-result all-vs-all gapped local alignment.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs alignall <i:sequenceDB> <i:resultDB> <o:alignmentDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 10.2.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `-a` | Add backtrace string (convert to alignments with mmseqs convertalis module) |
| `--alignment-mode` | How to compute the alignment: |
| `-e` | List matches below this E-value (range 0.0-inf) |
| `--min-seq-id` | List matches above this sequence identity (for clustering) (range 0.0-1.0) |
| `--min-aln-len` | Minimum alignment length (range 0-INT_MAX) |

## 10.3 `alignbykmer`

Heuristic gapped local k-mer based alignment.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs alignbykmer <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 10.3.1 Key Options

| Option | Purpose |
|---|---|
| `-k` | k-mer length (0: automatically set to optimum) |
| `--spaced-kmer-mode` | 0: use consecutive positions in k-mers; 1: use spaced k-mers |
| `--spaced-kmer-pattern` | User-specified spaced k-mer pattern |
| `--alph-size` | Alphabet size (range 2-21) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `-c` | List matches above this fraction of aligned (covered) residues (see –cov-mode) |
| `-e` | List matches below this E-value (range 0.0-inf) |
| `--cov-mode` | 0: coverage of query and target |

## 10.4 `expandaln`

Expand an alignment result based on another.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs expandaln <i:queryDB> <i:targetDB> <i:resultDB> <i:resultDB\|ca3mDB> <o:alignmentDB> [options]` |
| API layer | `mid_level_api` |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_PROFILE_PROFILE` |
| Called by modules | `search` |
| Calls modules | `n/a` |
| Related functional groups | `search_workflows` |
| Workflow script usage | `enrich.sh`, `iterativepp.sh` |

Reference links: Full CLI, Dependency map.

### 10.4.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--gap-open` | Gap open cost |
| `--gap-extend` | Gap extension cost |
| `--score-bias` | Score bias when computing SW alignment (in bits) |
| `-e` | List matches below this E-value (range 0.0-inf) |
| `--min-seq-id` | List matches above this sequence identity (for clustering) (range 0.0-1.0) |
| `-c` | List matches above this fraction of aligned (covered) residues (see –cov-mode) |

## 10.5 `fwbw`

Forward Backward Alignment.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |

| Aspect | Value |
|---|---|
| Workflow script usage | n/a |

Reference links: Full CLI, Dependency map.

### 10.6 `offsetalignment`

Offset alignment by ORF start position.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs offsetalignment <i:queryDB> <i:queryOrfDB> <i:targetDB> <i:targetOrfDB> <i:alnDB> <o:alnDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `cluster`, `linsearch`, `search` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `search_workflows` |
| Workflow script usage | `blastn.sh`, `linsearch.sh`, `nucleotide_clustering.sh`, `translated_search.sh` |

Reference links: Full CLI, Dependency map.

#### 10.6.1 Key Options

| Option | Purpose |
|---|---|
| `--search-type` | Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `--db-load-mode` | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| `--chain-alignments` | Chain overlapping alignments |
| `--merge-query` | Combine ORFs/split sequences to a single entry |

## 10.7 `proteinaln2nucl`

Transform protein alignments to nucleotide alignments.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs proteinaln2nucl <i:nuclQueryDB> <i:nuclTargetDB> <i:aaQueryDB> <i:aaTargetDB> <i:alnDB> <o:alnDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 10.7.1 Key Options

| Option | Purpose |
|---|---|
| `--gap-open` | Gap open cost |
| `--gap-extend` | Gap extension cost |
| `--sub-mat` | Substitution matrix file |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 10.8 `rescorediagonal`

Compute sequence identity for diagonal.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs rescorediagonal <i:queryDB> <i:targetDB> <i:prefilterDB> <o:resultDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Called by modules | `cluster, linclust, linsearch, search, taxonomy` |

| Aspect | Value |
|---|---|
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `search_workflows`, `taxonomy` |
| Workflow script usage | `linclust.sh`, `linsearch.sh`, `nucleotide_clustering.sh`, `taxpercontig.sh` |

Reference links: Full CLI, Dependency map.

### 10.8.1 Key Options

| Option | Purpose |
|---|---|
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--wrapped-scoring` | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| `-e` | List matches below this E-value (range 0.0-inf) |
| `-c` | List matches above this fraction of aligned (covered) residues (see –cov-mode) |
| `-a` | Add backtrace string (convert to alignments with mmseqs convertalis module) |
| `--cov-mode` | 0: coverage of query and target |
| `--min-seq-id` | List matches above this sequence identity (for clustering) (range 0.0-1.0) |
| `--min-aln-len` | Minimum alignment length (range 0-INT_MAX) |

## 10.9 `transitivealign`

Transfer alignments via transitivity.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |

| Aspect | Value |
|---|---|
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

# 11 Profiles

Modules for profile/MSA conversion, profile construction, and profile-driven workflow components.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

## 11.1 `convertca3m`

Convert a cA3M DB to a result DB.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_PROFILE_PROFILE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 11.2 `convertmsa`

Convert Stockholm/PFAM MSA file to a MSA DB.

| Aspect | Value |
|---|---|
| Usage | `usage:`<br>`mmseqs convertmsa <i:msaFile.sto[.gz]>`<br>`<o:msaDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION` |

| Aspect | Value |
|---|---|
| Called by modules | `databases` |
| Calls modules | `n/a` |
| Related functional groups | `database` |
| Workflow script usage | `databases.sh` |

Reference links: Full CLI, Dependency map.

### 11.2.1 Key Options

| Option | Purpose |
|---|---|
| `--identifier-field` | Field from STOCKHOLM comments for choosing the MSA identifier: 0: ID, 1: AC. If the respective comment does not exist, the name of the first sequence will become the identifier |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 11.3 `convertprofiledb`

Convert a HH-suite HHM DB to a profile DB.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 11.4 `expand2profile`

Expand an alignment result based on another and create a profile.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |

| Aspect | Value |
|---|---|
| API layer | mid_level_api |
| Category flags | COMMAND_PROFILE_PROFILE |
| Called by modules | search |
| Calls modules | n/a |
| Related functional groups | search_workflows |
| Workflow script usage | iterativepp.sh |

Reference links: Full CLI, Dependency map.

### 11.5 msa2profile

Convert a MSA DB to a profile DB.

| Aspect | Value |
|---|---|
| Usage | usage: mmseqs msa2profile <i:msaDB> <o:profileDB> [options] |
| API layer | low_level_api |
| Category flags | COMMAND_PROFILE \| COMMAND_DATABASE_CREATION |
| Called by modules | databases, pickconsensusrep |
| Calls modules | n/a |
| Related functional groups | clustering, database |
| Workflow script usage | databases.sh, pickconsensusrep.sh |

Reference links: Full CLI, Dependency map.

#### 11.5.1 Key Options

| Option | Purpose |
|---|---|
| --comp-bias-corr | Correct for locally biased amino acid composition (range 0-1) |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |
| --gap-open | Gap open cost |
| --gap-extend | Gap extension cost |
| --match-mode | 0: Columns that have a residue in the first sequence are kept, 1: columns that have a |

| Option | Purpose |
|---|---|
| | residue in –match-ratio of all sequences are kept |
| `--match-ratio` | Columns that have a residue in this ratio of all sequences are kept |
| `--pseudo-cnt-mode` | use 0: substitution-matrix or 1: context-specific pseudocounts |
| `--pca` | Pseudo count admixture strength |

## 11.6 `msa2result`

Convert a MSA DB to a profile DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs msa2result <i:msaDB> <o:seqDB> <o:profileDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE \| COMMAND_EXPERT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 11.6.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--gap-open` | Gap open cost |
| `--gap-extend` | Gap extension cost |
| `--match-mode` | 0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in –match-ratio of all sequences are kept |

| Option | Purpose |
| --- | --- |
| `--match-ratio` | Columns that have a residue in this ratio of all sequences are kept |
| `--pseudo-cnt-mode` | use 0: substitution-matrix or 1: context-specific pseudocounts |
| `--pca` | Pseudo count admixture strength |

## 11.7 `pairaln`

Pair sequences to match best protein A and B from a species.

| Aspect | Value |
| --- | --- |
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_EXPERT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 11.8 `profile2consensus`

Extract consensus sequence DB from a profile DB.

| Aspect | Value |
| --- | --- |
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Called by modules | `search` |
| Calls modules | `n/a` |
| Related functional groups | `search_workflows` |
| Workflow script usage | `iterativepp.sh` |

Reference links: Full CLI, Dependency map.

## 11.9 `profile2neff`

Convert a profile DB to a tab-separated list of Neff scores.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 11.10 `profile2pssm`

Convert a profile DB to a tab-separated PSSM file.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 11.11 `profile2repseq`

Extract representative sequence DB from a profile DB.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 11.12 `result2profile`

Compute profile DB from a result DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs result2profile <i:queryDB> <i:targetDB> <i:resultDB> <o:profileDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Called by modules | `search` |
| Calls modules | `n/a` |
| Related functional groups | `search_workflows` |
| Workflow script usage | `blastpgp.sh`, `enrich.sh` |

Reference links: Full CLI, Dependency map.

### 11.12.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `-e` | List matches below this E-value (range 0.0-inf) |
| `--gap-open` | Gap open cost |
| `--gap-extend` | Gap extension cost |
| `--mask-profile` | Mask query sequence of profile using tantan [0,1] |
| `--e-profile` | Include sequences matches with < E-value thr. into the profile (>=0.0) |
| `--wg` | Use global sequence weighting for profile calculation |

## 11.13 `sequence2profile`

Turn sequence into profile by adding context specific pseudo counts.

| Aspect | Value |
| --- | --- |
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 11.14 `tsv2exprofiledb`

Create a expandable profile db from TSV files.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs tsv2exprofiledb <i:tsvFilesBase> <o:exprofileDB> [options]` |
| API layer | `mid_level_api` |
| Category flags | `COMMAND_PROFILE_PROFILE` |
| Called by modules | `n/a` |
| Calls modules | `aliasdb`, `compress`, `mvdb`, `rmdb`, `tsv2db` |
| Related functional groups | `database`, `utilities` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 11.14.1 Key Options

| Option | Purpose |
| --- | --- |
| `--gpu` | Use GPU (CUDA) if possible |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12 Database Management

Modules for creating, indexing, splitting, merging, and maintaining MMseqs2 database artifacts.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

## 12.1 `aliasdb`

Create relative symlink of DB to another name in the same folder.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs aliasdb <i:srcDB> <o:dstDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `tsv2exprofiledb` |
| Calls modules | `n/a` |
| Related functional groups | `profiles` |
| Workflow script usage | `tsv2exprofiledb.sh` |

Reference links: Full CLI, Dependency map.

### 12.1.1 Key Options

| Option | Purpose |
| --- | --- |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12.2 `concatdbs`

Concatenate two DBs, giving new IDs to entries from 2nd DB.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs concatdbs <i:DB> <i:DB> <o:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SET` |
| Called by modules | `cluster`, `clusterupdate`, `linsearch` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `search_workflows` |

| Aspect | Value |
| --- | --- |
| Workflow script usage | `linsearch.sh`, `nucleotide_clustering.sh`, `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

#### 12.2.1 Key Options

| Option | Purpose |
| --- | --- |
| `--preserve-keys` | The keys of the two DB should be distinct, and they will be preserved in the concatenation |
| `--take-larger-entry` | Only keep the larger entry (dataSize >) in the concatenation, both databases need the same keys in the index |
| `--compressed` | Write compressed output |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 12.3 `cpdb`

Copy a DB.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs cpdb <i:srcDB> <o:dstDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 12.3.1 Key Options

| Option | Purpose |
| --- | --- |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 12.4 `createdb`

Convert FASTA/Q file(s) to a sequence DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs createdb <i:fastaFile1[.gz\|.bz2]> ... <i:fastaFileN[.gz\|.bz2]>\|<i:stdin> <o:sequenceDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION` |
| Called by modules | `databases`, `easy-cluster`, `easy-linclust`, `easy-linsearch`, `easy-rbh`, `easy-search`, `easy-taxonomy`, `multihitdb` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows`, `multi_hit` |
| Workflow script usage | `databases.sh`, `easycluster.sh`, `easyrbh.sh`, `easysearch.sh`, `easytaxonomy.sh`, `multihitdb.sh` |

Reference links: Full CLI, Dependency map.

#### 12.4.1 Key Options

| Option | Purpose |
|---|---|
| `--dbtype` | Database type 0: auto, 1: amino acid 2: nucleotides |
| `--shuffle` | Shuffle input database |
| `--createdb-mode` | Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q) |
| `--id-offset` | Numeric ids in index file are offset by this value |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| `--write-lookup` | write .lookup file containing mapping from internal id, fasta id and file number |

## 12.5 `createindex`

Store precomputed index on disk to reduce search overhead.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs createindex <i:sequenceDB> <tmpDir> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION` |
| Called by modules | `n/a` |
| Calls modules | `extractframes`, `extractorfs`, `rmdb`, `splitsequence` |
| Related functional groups | `sequence_manipulation` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 12.5.1 Key Options

| Option | Purpose |
| --- | --- |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--alph-size` | Alphabet size (range 2-21) |
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--max-seqs` | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| `--mask` | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| `--mask-prob` | Mask sequences is probablity is above threshold |

## 12.6 `createlinindex`

Create linsearch index.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs createlinindex <i:sequenceDB> <tmpDir> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION \| COMMAND_EXPERT` |
| Called by modules | `easy-linsearch`, `easy-search` |
| Calls modules | `extractframes`, `extractorfs`, `rmdb`, `splitsequence` |
| Related functional groups | `easy_workflows`, `sequence_manipulation` |
| Workflow script usage | `easysearch.sh` |

Reference links: Full CLI, Dependency map.

### 12.6.1 Key Options

| Option | Purpose |
| --- | --- |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--split-memory-limit` | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| `--alph-size` | Alphabet size (range 2-21) |
| `--mask` | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| `--mask-prob` | Mask sequences is probablity is above threshold |
| `--mask-lower-case` | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region |
| `--mask-n-repeat` | Repeat letters that occure > threshold in a rwo |

## 12.7 `createsubdb`

Create a subset of a DB from list of DB keys.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs createsubdb <i:subsetFile\|DB> <i:DB> <o:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SET` |
| Called by modules | `cluster`, `clusterupdate`, `linclust`, `search`, `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `search_workflows`, `taxonomy` |
| Workflow script usage | `blastp.sh`, `cascaded_clustering.sh`, `clustering.sh`, `linclust.sh`, `nucleotide_clustering.sh`, `taxpercontig.sh`, `translated_search.sh`, `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

#### 12.7.1 Key Options

| Option | Purpose |
|---|---|
| `--subdb-mode` | Subdb mode 0: copy data 1: soft link data and write index |
| `--id-mode` | Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 12.8 `databases`

List and download databases.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION` |
| Called by modules | `n/a` |

| Aspect | Value |
|---|---|
| Calls modules | `convertmsa`, `createdb`, `createtaxdb`, `msa2profile`, `nrtotaxmapping`, `prefixid`, `rmdb`, `tar2db` |
| Related functional groups | `profiles`, `taxonomy`, `utilities` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 12.9 `db2tar`

Archive contents of a DB to a tar archive.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs db2tar <i:DB> <o:tar[.gz]> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION \| COMMAND_EXPERT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 12.9.1 Key Options

| Option | Purpose |
|---|---|
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12.10 `lndb`

Symlink a DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs lndb <i:srcDB> <o:dstDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |

| Aspect | Value |
|---|---|
| Workflow script usage | n/a |

Reference links: Full CLI, Dependency map.

#### 12.10.1 Key Options

| Option | Purpose |
|---|---|
| -v | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 12.11 mergedbs

Merge entries from multiple DBs.

| Aspect | Value |
|---|---|
| Usage | usage: mmseqs mergedbs <i:DB> <o:DB> <i:DB1> ... <i:DBn> [options] |
| API layer | low_level_api |
| Category flags | COMMAND_SET |
| Called by modules | cluster, clusterupdate, rbh, search |
| Calls modules | n/a |
| Related functional groups | clustering, search_workflows |
| Workflow script usage | blastp.sh, blastpgp.sh, cascaded_clustering.sh, enrich.sh, iterativepp.sh, rbh.sh, searchslicedtargetprofile.sh, update_clustering.sh |

Reference links: Full CLI, Dependency map.

#### 12.11.1 Key Options

| Option | Purpose |
|---|---|
| --compressed | Write compressed output |
| -v | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| --prefixes | Comma separated list of prefixes for each entry |
| --merge-stop-empty | Don't continue merging entries after an empty entry |

## 12.12 `mvdb`

Move a DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs mvdb <i:srcDB> <o:dstDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `cluster`, `clusterupdate`, `search`, `taxonomy`, `tsv2exprofiledb` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `profiles`, `search_workflows`, `taxonomy` |
| Workflow script usage | `blastp.sh`, `cascaded_clustering.sh`, `searchslicedtargetprofile.sh`, `taxonomy.sh`, `tsv2exprofiledb.sh`, `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 12.12.1 Key Options

| Option | Purpose |
|---|---|
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12.13 `renamedbkeys`

Create a new DB with original keys renamed.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs renamedbkeys <i:idMapFile\|stdin> <i:DB> <o:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Called by modules | `clusterupdate`, `pickconsensusrep` |
| Calls modules | `n/a` |
| Related functional groups | `clustering` |
| Workflow script usage | `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 12.13.1 Key Options

| Option | Purpose |
|---|---|
| `--subdb-mode` | Subdb mode 0: copy data 1: soft link data and write index |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12.14 `rmdb`

Remove a DB.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `cluster`, `clusterupdate`, `createindex`, `createlinindex`, `databases`, `easy-cluster`, `easy-linclust`, `easy-linsearch`, `easy-rbh`, `easy-search`, `easy-taxonomy`, `linclust`, `linsearch`, `multihitsearch`, `pickconsensusrep`, `rbh`, `search`, `taxonomy`, `tsv2exprofiledb` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `easy_workflows`, `multi_hit`, `profiles`, `search_workflows`, `taxonomy` |
| Workflow script usage | `blastn.sh`, `blastp.sh`, `blastpgp.sh`, `cascaded_clustering.sh`, `clustering.sh`, `createindex.sh`, `databases.sh`, `easycluster.sh`, `easyrbh.sh`, `easysearch.sh`, `easytaxonomy.sh`, `iterativepp.sh`, `linclust.sh`, `linsearch.sh`, `multihitsearch.sh`, `nucleotide_clustering.sh`, `pickconsensusrep.sh`, `rbh.sh`, `searchslicedtargetprofile.sh`, `searchtargetprofile.sh`, `taxonomy.sh`, `taxpercontig.sh`, `translated_search.sh`, |

| Aspect | Value |
| --- | --- |
| | tsv2exprofiledb.sh, update_clustering.sh |

Reference links: Full CLI, Dependency map.

## 12.15 splitdb

Split DB into subsets.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs splitdb <i:DB> <o:DB> [options]` |
| API layer | low_level_api |
| Category flags | COMMAND_SET |
| Called by modules | n/a |
| Calls modules | n/a |
| Related functional groups | n/a |
| Workflow script usage | n/a |

Reference links: Full CLI, Dependency map.

### 12.15.1 Key Options

| Option | Purpose |
| --- | --- |
| --split | Split input into N equally distributed chunks |
| --split-aa | Try to find the best split boundaries by entry lengths |
| --compressed | Write compressed output |
| -v | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12.16 splitsequence

Split sequences by length.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs splitsequence <i:sequenceDB> <o:sequenceDB> [options]` |
| API layer | low_level_api |
| Category flags | COMMAND_SEQUENCE |

| Aspect | Value |
| --- | --- |
| Called by modules | `createindex`, `createlinindex`, `search` |
| Calls modules | `n/a` |
| Related functional groups | `search_workflows` |
| Workflow script usage | `blastn.sh`, `createindex.sh` |

Reference links: Full CLI, Dependency map.

### 12.16.1 Key Options

| Option | Purpose |
| --- | --- |
| `--sequence-overlap` | Overlap between sequences |
| `--sequence-split-mode` | Sequence split mode 0: copy data, 1: soft link data and write new index, |
| `--headers-split-mode` | Header split mode: 0: split position, 1: original header |
| `--max-seq-len` | Maximum sequence length |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| `--create-lookup` | Create database lookup file (can be very large) |

## 12.17 `subtractdbs`

Remove all entries from first DB occurring in second DB by key.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs subtractdbs <i:resultDBLeft> <i:resultDBRight> <o:resultDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SET` |
| Called by modules | `cluster`, `search` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `search_workflows` |

| Aspect | Value |
|---|---|
| Workflow script usage | `blastpgp.sh`, `cascaded_clustering.sh`, `enrich.sh`, `iterativepp.sh`, `nucleotide_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 12.17.1 Key Options

| Option | Purpose |
|---|---|
| `-e` | List matches below this E-value (range 0.0-inf) |
| `--e-profile` | Include sequences matches with < E-value thr. into the profile (>=0.0) |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12.18 `swapdb`

Transpose DB with integer values in first column.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs swapdb <i:resultDB> <o:resultDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Called by modules | `cluster`, `clusterupdate`, `multihitdb`, `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `multi_hit`, `taxonomy` |
| Workflow script usage | `cascaded_clustering.sh`, `multihitdb.sh`, `taxpercontig.sh`, `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 12.18.1 Key Options

| Option | Purpose |
| --- | --- |
| `--split-memory-limit` | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12.19 `tar2db`

Convert content of tar archives to any DB.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs tar2db <i:tar[.gz]> ... <i:tar[.gz]> <o:resultDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION \| COMMAND_EXPERT` |
| Called by modules | `databases` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `databases.sh` |

Reference links: Full CLI, Dependency map.

### 12.19.1 Key Options

| Option | Purpose |
| --- | --- |
| `--output-dbtype` | Set database type for resulting database: Amino acid sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6, Prefiltering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15 |
| `--tar-include` | Include file names based on this regex |
| `--tar-exclude` | Exclude file names based on this regex |

| Option | Purpose |
|---|---|
| `--compressed` | Write compressed output |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 12.20 `tsv2db`

Convert a TSV file to any DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs tsv2db <i:tsvFile> <o:resultDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION \| COMMAND_EXPERT` |
| Called by modules | `cluster`, `multihitdb`, `pickconsensusrep`, `tsv2exprofiledb` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `multi_hit`, `profiles` |
| Workflow script usage | `cascaded_clustering.sh`, `multihitdb.sh`, `pickconsensusrep.sh`, `tsv2exprofiledb.sh` |

Reference links: Full CLI, Dependency map.

### 12.20.1 Key Options

| Option | Purpose |
|---|---|
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--output-dbtype` | Set database type for resulting database: Amino acid sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6, Prefiltering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15 |
| `--compressed` | Write compressed output |

| Option | Purpose |
|---|---|
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

# 13 Sequence Manipulation

Modules that transform sequence content, frames, ORFs, and masked/aligned regions.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

> **Warning**
>
> Validate database-type and sidecar compatibility before chaining modules. Most pipeline failures come from DB contract mismatches.

## 13.1 `extractalignedregion`

Extract aligned sequence region from query.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs extractalignedregion <i:queryDB> <i:targetDB> <i:resultDB> <o:sequenceDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 13.1.1 Key Options

| Option | Purpose |
|---|---|
| `--extract-mode` | Extract from 1: Query, 2: Target |
| `--compressed` | Write compressed output |

| Option | Purpose |
|---|---|
| `--db-load-mode` | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 13.2 `extractframes`

Extract frames from a nucleotide sequence DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs extractframes <i:sequenceDB> <o:sequenceDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Called by modules | `cluster`, `createindex`, `createlinindex`, `search` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `database`, `search_workflows` |
| Workflow script usage | `blastn.sh`, `createindex.sh`, `nucleotide_clustering.sh`, `translated_search.sh` |

Reference links: Full CLI, Dependency map.

### 13.2.1 Key Options

| Option | Purpose |
|---|---|
| `--forward-frames` | Comma-separated list of frames on the forward strand to be extracted |
| `--reverse-frames` | Comma-separated list of frames on the reverse strand to be extracted |
| `--translation-table` | 1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE |
| `--translate` | Translate ORF to amino acid |

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| `--create-lookup` | Create database lookup file (can be very large) |

## 13.3 `extractorfs`

Six-frame extraction of open reading frames.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs extractorfs <i:sequenceDB> <o:sequenceDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Called by modules | `createindex`, `createlinindex`, `linsearch`, `multihitdb`, `search`, `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `database`, `multi_hit`, `search_workflows`, `taxonomy` |
| Workflow script usage | `createindex.sh`, `multihitdb.sh`, `taxpercontig.sh`, `translated_search.sh` |

Reference links: Full CLI, Dependency map.

### 13.3.1 Key Options

| Option | Purpose |
|---|---|
| `--min-length` | Minimum codon number in open reading frames |
| `--max-length` | Maximum codon number in open reading frames |
| `--max-gaps` | Maximum number of codons with gaps or unknown residues before an open reading frame is rejected |
| `--contig-start-mode` | Contig start can be 0: incomplete, 1: complete, 2: both |

| Option | Purpose |
|---|---|
| `--contig-end-mode` | Contig end can be 0: incomplete, 1: complete, 2: both |
| `--orf-start-mode` | Orf fragment can be 0: from start to stop, 1: from any to stop, 2: from last encountered start to stop (no start in the middle) |
| `--forward-frames` | Comma-separated list of frames on the forward strand to be extracted |
| `--reverse-frames` | Comma-separated list of frames on the reverse strand to be extracted |

## 13.4 `masksequence`

Soft mask sequence DB using tantan.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs masksequence <i:sequenceDB> <o:sequenceDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 13.4.1 Key Options

| Option | Purpose |
|---|---|
| `--mask` | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| `--mask-prob` | Mask sequences is probablity is above threshold |
| `--mask-lower-case` | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region |
| `--mask-n-repeat` | Repeat letters that occure > threshold in a rwo |
| `--threads` | Number of CPU-cores used (all by default) |

| Option | Purpose |
|---|---|
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 13.5 `orftocontig`

Write ORF locations in alignment format.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs orftocontig <i:contigsSequenceDB> <i:extractedOrfsHeadersDB> <o:orfsAlignedToContigDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Called by modules | `multihitdb` |
| Calls modules | `n/a` |
| Related functional groups | `multi_hit` |
| Workflow script usage | `multihitdb.sh` |

Reference links: Full CLI, Dependency map.

### 13.5.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 13.6 `recoverlongestorf`

Recover longest ORF for taxonomy annotation after elimination.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs recoverlongestorf <i:orfDB> <i:resultDB> <o:tsvFile> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_EXPERT` |

| Aspect | Value |
|---|---|
| Called by modules | `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `taxonomy` |
| Workflow script usage | `taxpercontig.sh` |

Reference links: Full CLI, Dependency map.

### 13.6.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 13.7 `reverseseq`

Reverse (without complement) sequences.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs reverseseq <i:sequenceDB> <o:revSequenceDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 13.7.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 13.8 `translateaa`

Translate proteins to lexicographically lowest codons.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs translateaa <i:sequenceDB> <o:sequenceDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 13.8.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 13.9 `translatenucs`

Translate nucleotides to proteins.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs translatenucs <i:sequenceDB> <o:sequenceDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Called by modules | `multihitdb` |
| Calls modules | `n/a` |
| Related functional groups | `multi_hit` |
| Workflow script usage | `multihitdb.sh` |

Reference links: Full CLI, Dependency map.

### 13.9.1 Key Options

| Option | Purpose |
|---|---|
| `--translation-table` | 1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE |
| `--add-orf-stop` | Add stop codon '*' at complete start and end |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| `--compressed` | Write compressed output |
| `--threads` | Number of CPU-cores used (all by default) |

# 14 Result Handling

Modules that transform, filter, summarize, and export result databases.

**Note**

This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

**Warning**

Validate database-type and sidecar compatibility before chaining modules. Most pipeline failures come from DB contract mismatches.

## 14.1 `convert2fasta`

Convert sequence DB to FASTA format.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |

| Aspect | Value |
|---|---|
| Workflow script usage | n/a |

Reference links: Full CLI, Dependency map.

## 14.2 convertalis

Convert alignment DB to BLAST-tab, SAM or custom format.

| Aspect | Value |
|---|---|
| Usage | usage: mmseqs convertalis <i:queryDb> <i:targetDb> <i:alignmentDB> <o:alignmentFile> [options] |
| API layer | low_level_api |
| Category flags | COMMAND_FORMAT_CONVERSION |
| Called by modules | easy-linsearch, easy-rbh, easy-search, easy-taxonomy |
| Calls modules | n/a |
| Related functional groups | easy_workflows |
| Workflow script usage | easyrbh.sh, easysearch.sh, easytaxonomy.sh |

Reference links: Full CLI, Dependency map.

### 14.2.1 Key Options

| Option | Purpose |
|---|---|
| --gap-open | Gap open cost |
| --gap-extend | Gap extension cost |
| --format-mode | Output format: |
| --format-output | Choose comma separated list of output columns from: query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen |
| --translation-table | 1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE |

| Option | Purpose |
|---|---|
| `--search-type` | Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment |
| `--sub-mat` | Substitution matrix file |
| `--db-load-mode` | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |

## 14.3 `createseqfiledb`

Create a DB of unaligned FASTA entries.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs createseqfiledb <i:sequenceDB> <i:resultDB> <o:fastaDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION \| COMMAND_EXPERT` |
| Called by modules | `easy-cluster`, `easy-linclust` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows` |
| Workflow script usage | `easycluster.sh` |

Reference links: Full CLI, Dependency map.

### 14.3.1 Key Options

| Option | Purpose |
|---|---|
| `--min-sequences` | Minimum number of sequences a cluster may contain |
| `--max-sequences` | Maximum number of sequences a cluster may contain |
| `--hh-format` | Format entries to use with hhsuite (for singleton clusters) |
| `--db-load-mode` | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |

| Option | Purpose |
|---|---|
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 14.4 `createtsv`

Convert result DB to tab-separated flat file.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs createtsv <i:queryDB> [<i:targetDB>] <i:resultDB> <o:tsvFile> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION` |
| Called by modules | `easy-cluster`, `easy-linclust`, `easy-taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows` |
| Workflow script usage | `easycluster.sh`, `easytaxonomy.sh` |

Reference links: Full CLI, Dependency map.

#### 14.4.1 Key Options

| Option | Purpose |
|---|---|
| `--first-seq-as-repr` | Use the first sequence of the clustering result as representative sequence |
| `--target-column` | Select a target column (default 1), 0 if no target id exists |
| `--full-header` | Replace DB ID by its corresponding Full Header |
| `--idx-seq-src` | 0: auto, 1: split/translated sequences, 2: input sequences |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| `--db-output` | Return a result DB instead of a text file |

## 14.5 `extractdomains`

Extract highest scoring alignment regions for each sequence from BLAST-tab file.

| Aspect | Value |
| --- | --- |
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 14.6 `filterresult`

Pairwise alignment result filter.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs filterresult <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `search` |
| Calls modules | `n/a` |
| Related functional groups | `search_workflows` |
| Workflow script usage | `searchslicedtargetprofile.sh` |

Reference links: Full CLI, Dependency map.

### 14.6.1 Key Options

| Option | Purpose |
| --- | --- |
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | keep the query (representative) sequence |

| Option | Purpose |
|---|---|
| `--gap-open` | Gap open cost |
| `--gap-extend` | Gap extension cost |
| `--filter-min-enable` | Only filter MSAs with more than N sequences, 0 always filters |
| `--max-seq-id` | Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] |
| `--qid` | Reduce diversity of output MSAs using min.seq. identity with query sequences |

## 14.7 `result2dnamsa`

Compute MSA DB with out insertions in the query for DNA sequences.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs result2dnamsa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 14.7.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| `--skip-query` | Skip the query sequence |

## 14.8 `result2flat`

Create flat file by adding FASTA headers to DB entries.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs result2flat <i:queryDB> <i:targetDB> <i:resultDB> <o:fastaDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION \| COMMAND_EXPERT` |
| Called by modules | `easy-cluster`, `easy-linclust` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows` |
| Workflow script usage | `easycluster.sh` |

Reference links: Full CLI, Dependency map.

### 14.8.1 Key Options

| Option | Purpose |
|---|---|
| `--use-fasta-header` | Use the id parsed from the fasta header as the index key instead of using incrementing numeric identifiers |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 14.9 `result2msa`

Compute MSA DB from a result DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs result2msa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `pickconsensusrep` |
| Calls modules | `n/a` |
| Related functional groups | `clustering` |
| Workflow script usage | `pickconsensusrep.sh` |

Reference links: Full CLI, Dependency map.

### 14.9.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--gap-open` | Gap open cost |
| `--gap-extend` | Gap extension cost |
| `--filter-msa` | Filter msa: 0: do not filter, 1: filter |
| `--filter-min-enable` | Only filter MSAs with more than N sequences, 0 always filters |
| `--max-seq-id` | Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] |
| `--qid` | Reduce diversity of output MSAs using min.seq. identity with query sequences |

## 14.10 `result2rbh`

Filter a merged result DB to retain only reciprocal best hits.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs result2rbh <i:resultDB> <o:resultDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `rbh` |
| Calls modules | `n/a` |
| Related functional groups | `search_workflows` |
| Workflow script usage | `rbh.sh` |

Reference links: Full CLI, Dependency map.

### 14.10.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 14.11 result2repseq

Get representative sequences from result DB.

| Aspect | Value |
|---|---|
| Usage | usage: mmseqs result2repseq <i:sequenceDB> <i:resultDB> <o:sequenceDb> [options] |
| API layer | low_level_api |
| Category flags | COMMAND_RESULT |
| Called by modules | clusterupdate, easy-cluster, easy-linclust |
| Calls modules | n/a |
| Related functional groups | clustering, easy_workflows |
| Workflow script usage | easycluster.sh, update_clustering.sh |

Reference links: Full CLI, Dependency map.

### 14.11.1 Key Options

| Option | Purpose |
|---|---|
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --compressed | Write compressed output |
| --threads | Number of CPU-cores used (all by default) |
| -v | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 14.12 result2stats

Compute statistics for each entry in a DB.

| Aspect | Value |
|---|---|
| Usage | usage: mmseqs result2stats <i:queryDB> <i:targetDB> <i:resultDB> <o:statsDB> [options] |
| API layer | low_level_api |
| Category flags | COMMAND_RESULT |
| Called by modules | multihitdb, search |
| Calls modules | n/a |

| Aspect | Value |
|---|---|
| Related functional groups | `multi_hit`, `search_workflows` |
| Workflow script usage | `multihitdb.sh`, `searchslicedtargetprofile.sh` |

Reference links: Full CLI, Dependency map.

### 14.12.1 Key Options

| Option | Purpose |
|---|---|
| `--stat` | One of: linecount, mean, min, max, doolittle, charges, seqlen, firstline |
| `--tsv` | Return output in TSV format |
| `--compressed` | Write compressed output |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 14.13 `sortresult`

Sort a result DB in the same order as the prefilter or align module.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs sortresult <i:resultbDB> <o:resultDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 14.13.1 Key Options

| Option | Purpose |
|---|---|
| `--compressed` | Write compressed output |
| `--threads` | Number of CPU-cores used (all by default) |

| Option | Purpose |
|---|---|
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 14.14 `summarizealis`

Summarize alignment result to one row (uniq. cov., cov., avg. seq. id.).

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs summarizealis <i:alignmentDB> <o:summerizedDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `easy-taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows` |
| Workflow script usage | `easytaxonomy.sh` |

Reference links: Full CLI, Dependency map.

### 14.14.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 14.15 `summarizeheaders`

Summarize FASTA headers of result DB.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 14.16 `summarizeresult`

Extract annotations from alignment DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs summarizeresult <i:alignmentDB> <o:alignmentDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Called by modules | `easy-linsearch`, `easy-search` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows` |
| Workflow script usage | `easysearch.sh` |

Reference links: Full CLI, Dependency map.

### 14.16.1 Key Options

| Option | Purpose |
|---|---|
| `-a` | Add backtrace string (convert to alignments with mmseqs convertalis module) |
| `-c` | List matches above this fraction of aligned (covered) residues (see –cov-mode) |
| `--overlap` | Maximum overlap of covered regions |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 14.17 `swapresults`

Transpose prefilter/alignment DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs swapresults <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB> [options]` |
| API layer | `low_level_api` |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_RESULT` |
| Called by modules | `easy-taxonomy`, `linsearch`, `rbh`, `search` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows`, `search_workflows` |
| Workflow script usage | `easytaxonomy.sh`, `linsearch.sh`, `rbh.sh`, `searchslicedtargetprofile.sh`, `searchtargetprofile.sh` |

Reference links: Full CLI, Dependency map.

#### 14.17.1 Key Options

| Option | Purpose |
|---|---|
| `--split-memory-limit` | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| `-e` | List matches below this E-value (range 0.0-inf) |
| `--gap-open` | Gap open cost |
| `--gap-extend` | Gap extension cost |
| `--sub-mat` | Substitution matrix file |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `--db-load-mode` | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |

# 15 Multi-hit

Modules for grouped-sequence (set-based) search and per-set aggregation pipelines.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

## 15.1 `besthitperset`

For each set of sequences compute the best element and update p-value.

| Aspect | Value |
|---|---|
| Usage | `usage:`<br>`mmseqs besthitperset <i:targetSetDB> <i:resultDB> <o:resultDB> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Called by modules | `multihitsearch` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `multihitsearch.sh` |

Reference links: Full CLI, Dependency map.

### 15.1.1 Key Options

| Option | Purpose |
|---|---|
| `--simple-best-hit` | Update the p-value by a single best hit, or by best and second best hits |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 15.2 `combinepvalperset`

For each set compute the combined p-value.

| Aspect | Value |
|---|---|
| Usage | `usage:`<br>`mmseqs combinepvalperset <i:querySetDB> <i:targetSetDB> <i:resultDB> <o:pvalDB> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 15.2.1 Key Options

| Option | Purpose |
|---|---|
| `--alpha` | Set alpha for combining p-values during aggregation |
| `--aggregation-mode` | Combined P-values computed from 0: multihit, 1: minimum of all P-values, 2: product-of-P-values, 3: truncated product |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 15.3 `mergeresultsbyset`

Merge results from multiple ORFs back to their respective contig.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs mergeresultsbyset <i:setDB> <i:DB> <o:DB> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Called by modules | `multihitsearch`, `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `taxonomy` |
| Workflow script usage | `multihitsearch.sh`, `taxpercontig.sh` |

Reference links: Full CLI, Dependency map.

### 15.3.1 Key Options

| Option | Purpose |
|---|---|
| `--db-load-mode` | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 15.4 `multihitdb`

Create sequence DB for multi hit searches.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs multihitdb <i:fastaFile1[.gz\|bz2]> ... <i:fastaFileN[.gz\|bz2]> <o:setDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Called by modules | `n/a` |
| Calls modules | `createdb`, `extractorfs`, `filterdb`, `orftocontig`, `result2stats`, `swapdb`, `translatenucs`, `tsv2db` |
| Related functional groups | `database`, `result_handling`, `sequence_manipulation`, `utilities` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 15.4.1 Key Options

| Option | Purpose |
|---|---|
| `--dbtype` | Database type 0: auto, 1: amino acid 2: nucleotides |
| `--shuffle` | Shuffle input database |
| `--createdb-mode` | Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q) |
| `--id-offset` | Numeric ids in index file are offset by this value |
| `--min-length` | Minimum codon number in open reading frames |
| `--max-length` | Maximum codon number in open reading frames |
| `--max-gaps` | Maximum number of codons with gaps or unknown residues before an open reading frame is rejected |

| Option | Purpose |
|---|---|
| `--contig-start-mode` | Contig start can be 0: incomplete, 1: complete, 2: both |

### 15.5 `multihitsearch`

Search with a grouped set of sequences against another grouped set.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs multihitsearch <i:querySetDB> <i:targetSetDB> <o:resultDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Called by modules | `n/a` |
| Calls modules | `besthitperset`, `mergeresultsbyset`, `rmdb`, `search` |
| Related functional groups | `database`, `search_workflows` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 15.5.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |

| Option | Purpose |
|---|---|
| `--k-score` | k-mer threshold for generating similar k-mer lists |

# 16 Taxonomy

Modules for taxonomy DB preparation, assignment, filtering, and reporting workflows.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

> **Warning**
>
> Validate database-type and sidecar compatibility before chaining modules. Most pipeline failures come from DB contract mismatches.

## 16.1 `addtaxonomy`

Add taxonomic labels to result DB.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Called by modules | `easy-taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows` |
| Workflow script usage | `easytaxonomy.sh` |

Reference links: Full CLI, Dependency map.

## 16.2 `aggregatetax`

Aggregate multiple taxon labels to a single label.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_TAXONOMY` |
| Called by modules | `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 16.3 `aggregatetaxweights`

Aggregate multiple taxon labels to a single label.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Called by modules | `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `taxpercontig.sh` |

Reference links: Full CLI, Dependency map.

### 16.4 `createbintaxmapping`

Create binary taxonomy mapping from tabular taxonomy mapping.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 16.5 `createbintaxonomy`

Create binary taxonomy from NCBI input.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Called by modules | `createtaxdb` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `createtaxdb.sh` |

Reference links: Full CLI, Dependency map.

### 16.6 `createdmptaxonomy`

Create dmp files from binary taxonomy.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 16.7 `createtaxdb`

Add taxonomic labels to sequence DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs createtaxdb <i:sequenceDB> <tmpDir> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Called by modules | `databases` |

| Aspect | Value |
|---|---|
| Calls modules | `createbintaxonomy` |
| Related functional groups | `database` |
| Workflow script usage | `databases.sh` |

Reference links: Full CLI, Dependency map.

### 16.7.1 Key Options

| Option | Purpose |
|---|---|
| `--ncbi-tax-dump` | tax dump directory. The tax dump can be downloaded here "ftp://ftp.ncbi.nlm.nih.gov/pub/taxonomy/taxdump.tar.gz" |
| `--tax-mapping-file` | File to map sequence identifier to taxonomical identifier |
| `--tax-mapping-mode` | Map taxonomy based on sequence database 0: .lookup file 1: .source file |
| `--tax-db-mode` | Create taxonomy database as: 0: .dmp flat files (human readable) 1: binary dump (faster readin) |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 16.8 `filtertaxdb`

Filter taxonomy result database.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs filtertaxdb <i:targetDB> <i:taxDB> <o:taxDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 16.8.1 Key Options

| Option | Purpose |
|---|---|
| `--taxon-list` | Taxonomy ID, possibly multiple values separated by ',' |
| `--compressed` | Write compressed output |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 16.9 `filtertaxseqdb`

Filter taxonomy sequence database.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs filtertaxseqdb <i:taxSeqDB> <o:taxSeqDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 16.9.1 Key Options

| Option | Purpose |
|---|---|
| `--taxon-list` | Taxonomy ID, possibly multiple values separated by ',' |
| `--subdb-mode` | Subdb mode 0: copy data 1: soft link data and write index |
| `--compressed` | Write compressed output |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 16.10 `lca`

Compute the lowest common ancestor.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs lca <i:targetDB> <i:resultDB> <o:taxaDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Called by modules | `easy-taxonomy`, `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows` |
| Workflow script usage | `taxonomy.sh` |

Reference links: Full CLI, Dependency map.

### 16.10.1 Key Options

| Option | Purpose |
|---|---|
| `--lca-ranks` | Add column with specified ranks (',' separated) |
| `--blacklist` | Comma separated list of ignored taxa in LCA computation |
| `--tax-lineage` | 0: don't show, 1: add all lineage names, 2: add all lineage taxids |
| `--compressed` | Write compressed output |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 16.11 `lcaalign`

Efficient gapped alignment for lca computation.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs lcaalign <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Called by modules | `search` |
| Calls modules | `n/a` |

| Aspect | Value |
|---|---|
| Related functional groups | `search_workflows` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 16.11.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `-a` | Add backtrace string (convert to alignments with mmseqs convertalis module) |
| `--alignment-mode` | How to compute the alignment: |
| `--alignment-output-mode` | How to compute the alignment: |
| `--wrapped-scoring` | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| `-e` | List matches below this E-value (range 0.0-inf) |

### 16.12 `majoritylca`

Compute the lowest common ancestor using majority voting.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 16.13 `nrtotaxmapping`

Create taxonomy mapping for NR database.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `databases` |
| Calls modules | `n/a` |
| Related functional groups | `database` |
| Workflow script usage | `databases.sh` |

Reference links: Full CLI, Dependency map.

## 16.14 `taxonomy`

Taxonomic classification.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs taxonomy <i:queryDB> <i:targetDB> <o:taxaDB> <tmpDir> [options]` |
| API layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Called by modules | `easy-taxonomy`, `taxonomy` |
| Calls modules | `aggregatetax`, `aggregatetaxweights`, `createsubdb`, `extractorfs`, `filterdb`, `lca`, `mergeresultsbyset`, `mvdb`, `prefilter`, `recoverlongestorf`, `rescorediagonal`, `rmdb`, `search`, `swapdb`, `taxonomy` |
| Related functional groups | `alignment`, `database`, `easy_workflows`, `multi_hit`, `prefiltering`, `search_workflows`, `sequence_manipulation`, `utilities` |
| Workflow script usage | `easytaxonomy.sh`, `taxpercontig.sh` |

Reference links: Full CLI, Dependency map.

### 16.14.1 Key Options

| Option | Purpose |
|---|---|
| `--comp-bias-corr` | Correct for locally biased amino acid composition (range 0-1) |
| `--comp-bias-corr-scale` | Correct for locally biased amino acid composition (range 0-1) |
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--seed-sub-mat` | Substitution matrix file for k-mer generation |
| `-s` | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |
| `-k` | k-mer length (0: automatically set to optimum) |
| `--target-search-mode` | target search mode (0: regular k-mer, 1: similar k-mer) |
| `--k-score` | k-mer threshold for generating similar k-mer lists |

## 16.15 `taxonomyreport`

Create a taxonomy report in Kraken or Krona format.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs taxonomyreport <i:seqTaxDB> <i:taxResultDB/resultDB/sequenceDB> <o:taxonomyReport> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_FORMAT_CONVERSION` |
| Called by modules | `easy-taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `easy_workflows` |
| Workflow script usage | `easytaxonomy.sh` |

Reference links: Full CLI, Dependency map.

#### 16.15.1 Key Options

| Option | Purpose |
|---|---|
| `--report-mode` | Taxonomy report mode 0: Kraken 1: Krona |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

# 17 Utilities

General-purpose helpers and special-purpose modules that support advanced workflow composition.

> **Note**
>
> This page emphasizes module relationships and practical options. For complete CLI details, open the linked command reference pages. In connection tables, `n/a` means no direct static edge was resolved.

## 17.1 `apply`

Execute given program on each DB entry.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs apply <i:DB> <o:DB> --program [args...] [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 17.1.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 17.2 `compress`

Compress DB entries.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs compress <i:DB> <o:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `tsv2exprofiledb` |
| Calls modules | `n/a` |
| Related functional groups | `profiles` |
| Workflow script usage | `tsv2exprofiledb.sh` |

Reference links: Full CLI, Dependency map.

### 17.2.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 17.3 `convertkb`

Convert UniProtKB data to a DB.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 17.4 `decompress`

Decompress DB entries.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs decompress <i:DB> <o:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 17.4.1 Key Options

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 17.5 `diffseqdbs`

Compute diff of two sequence DBs.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `clusterupdate` |
| Calls modules | `n/a` |
| Related functional groups | `clustering` |
| Workflow script usage | `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

## 17.6 `filterdb`

DB filtering by given conditions.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs filterdb <i:resultDB> <o:resultDB> [options]` |

| Aspect | Value |
|---|---|
| API layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Called by modules | `cluster`, `clusterupdate`, `easy-taxonomy`, `linclust`, `linsearch`, `multihitdb`, `rbh`, `taxonomy` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `easy_workflows`, `multi_hit`, `search_workflows`, `taxonomy` |
| Workflow script usage | `cascaded_clustering.sh`, `easytaxonomy.sh`, `linclust.sh`, `linsearch.sh`, `multihitdb.sh`, `rbh.sh`, `taxonomy.sh`, `taxpercontig.sh`, `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

### 17.6.1 Key Options

| Option | Purpose |
|---|---|
| `--add-self-matches` | Artificially add entries of queries with themselves (for clustering) |
| `--filter-expression` | Specify a mathematical expression to filter lines |
| `--filter-column` | column |
| `--column-to-take` | column to take in join mode. If −1, the whole line is taken |
| `--filter-regex` | Regex to select column (example float: [0-9]*(.[0-9]+)? int:[1-9]{1}[0-9]) |
| `--positive-filter` | Used in conjunction with –filter-file. If true, out = in filter ; if false, out = in - filter |
| `--filter-file` | Specify a file that contains the filtering elements |
| `--beats-first` | Filter by comparing each entry to the first entry |

## 17.7 `gff2db`

Extract regions from a sequence database based on a GFF3 file.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

## 17.8 `gpuserver`

Start a GPU server.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs gpuserver <i:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 17.8.1 Key Options

| Option | Purpose |
|---|---|
| `--max-seqs` | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| `--prefilter-mode` | prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped |
| `--gpu` | Use GPU (CUDA) if possible |
| `--db-load-mode` | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |

### 17.9 `maskbygff`

Mask out sequence regions in a sequence DB by features selected from a GFF3 file.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 17.10 `prefixid`

For each entry in a DB prepend the entry key to the entry itself.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs prefixid <i:DB> <o:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Called by modules | `clusterupdate`, `databases`, `pickconsensusrep` |
| Calls modules | `n/a` |
| Related functional groups | `clustering`, `database` |
| Workflow script usage | `databases.sh`, `pickconsensusrep.sh`, `update_clustering.sh` |

Reference links: Full CLI, Dependency map.

#### 17.10.1 Key Options

| Option | Purpose |
|---|---|
| `--prefix` | Use this prefix for all entries |
| `--mapping-file` | Specify a file that translates the keys of a DB to new keys, TSV format |
| `--tsv` | Return output in TSV format |

| Option | Purpose |
|---|---|
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 17.11 `setextendeddbtype`

Write an extended DB.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs setextendeddbtype <i:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 17.11.1 Key Options

| Option | Purpose |
|---|---|
| `--extended-dbtype` | Set extended dbtype 1: compressed, 2: need src, 4: context pseudoe cnts |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 17.12 `suffixid`

For each entry in a DB append the entry key to the entry itself.

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs suffixid <i:resultDB> <o:resultDB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Called by modules | `n/a` |

| Aspect | Value |
|---|---|
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 17.12.1 Key Options

| Option | Purpose |
|---|---|
| `--prefix` | Use this prefix for all entries |
| `--mapping-file` | Specify a file that translates the keys of a DB to new keys, TSV format |
| `--tsv` | Return output in TSV format |
| `--threads` | Number of CPU-cores used (all by default) |
| `--compressed` | Write compressed output |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 17.13 `summarizetabs`

Extract annotations from HHblits BLAST-tab-formatted results.

| Aspect | Value |
|---|---|
| Usage | Help snapshot unavailable locally. |
| API layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 17.14 `touchdb`

Preload DB into memory (page cache).

| Aspect | Value |
|---|---|
| Usage | `usage: mmseqs touchdb <i:DB> [options]` |

| Aspect | Value |
| --- | --- |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 17.14.1 Key Options

| Option | Purpose |
| --- | --- |
| `--threads` | Number of CPU-cores used (all by default) |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 17.15 `unpackdb`

Unpack a DB into separate files.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs unpackdb <i:DB> <o:outDir> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

### 17.15.1 Key Options

| Option | Purpose |
| --- | --- |
| `--unpack-name-mode` | Name unpacked files by 0: DB key, 1: accession (through .lookup) |
| `--unpack-suffix` | File suffix for unpacked files. |
| `--threads` | Number of CPU-cores used (all by default) |

| Option | Purpose |
| --- | --- |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 17.16 `view`

Print DB entries given in –id-list to stdout.

| Aspect | Value |
| --- | --- |
| Usage | `usage: mmseqs view <i:DB> [options]` |
| API layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Called by modules | `n/a` |
| Calls modules | `n/a` |
| Related functional groups | `n/a` |
| Workflow script usage | `n/a` |

Reference links: Full CLI, Dependency map.

#### 17.16.1 Key Options

| Option | Purpose |
| --- | --- |
| `--id-list` | Entries to be printed separated by ‘,’ |
| `--id-mode` | Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup) |
| `--idx-entry-type` | 0: sequence, 1: src sequence, 2: header, 3: src header |
| `-v` | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 18 Expert Manual

This chapter focuses on advanced pipeline composition, reproducibility controls, and documentation maintenance. Foundational storage and parallel mechanics are summarized in `foundations.md`; this chapter assumes that baseline model.

### 18.1 1. Expert Operating Discipline

Custom MMseqs2 pipelines fail most often from implicit assumptions, not syntax errors. Advanced users should lock assumptions explicitly before scaling.

| Discipline | Practical Rule |
|---|---|
| Contract discipline | Validate DB type and sidecar completeness at each boundary |
| Mode discipline | Keep alignment, rescoring, and filter modes fixed across comparisons |
| Topology discipline | Confirm real call path with dependency map before debugging behavior |
| Scale discipline | Validate on representative subsets before full runs |

## 18.2 2. Contract Enforcement in Custom Pipelines

Module chains should be reviewed as explicit contracts:

| Contract Surface | What to Verify |
|---|---|
| Structural contract | `.dbtype` and index compatibility with downstream module requirements |
| Metadata contract | Header, lookup, taxonomy, and mapping sidecars required by downstream exports |
| Semantic contract | Output fields remain meaningful under selected mode and filter policy |

A common anti-pattern is tuning thresholds while the contract itself is broken. In practice, contract validation should happen before parameter optimization.

> **Warning**
>
> If a downstream result looks wrong, verify contract and mode assumptions first. Parameter tuning should be the last step.

## 18.3 3. Reproducibility Controls

Reproducible comparisons need a stable run envelope, not only stable command names.

| Reproducibility Domain | Controls to Pin |
|---|---|
| Core scoring semantics | `--alignment-mode`, `--alignment-output-mode`, `--rescore-mode` |
| Selection semantics | `-e`, `-c`, `--cov-mode`, `--max-accept`, `--max-rejected` |
| Profile behavior | Pseudocount, weighting, and MSA filtering settings |

| Reproducibility Domain | Controls to Pin |
|---|---|
| Taxonomy behavior | ORF/frame extraction policy and aggregation/report mode |
| Runtime envelope | Split policy, load mode, thread/MPI strategy, temporary-storage topology |

When benchmarking, record these controls as part of run metadata. Without this context, result deltas are difficult to interpret.

## 18.4 4. Performance Triage for Advanced Workloads

For slow or unstable large runs, triage in this order:

| Step | Question |
|---|---|
| 1 | Is startup dominated by index read-in or shared-storage contention? |
| 2 | Is split policy over-reducing memory at the cost of merge and I/O overhead? |
| 3 | Is the distributed mode aligned with actual storage topology? |
| 4 | Are mode and filter choices inflating downstream data volume unexpectedly? |

This sequence is intentionally infrastructure-first. It reflects how MMseqs2 runtime is typically determined at scale.

> **Performance**
>
> In most production pipelines, index/load/split decisions create larger runtime swings than final threshold adjustments.

## 18.5 5. Documentation Maintenance Workflow

Use this workflow when MMseqs2 source, options, or dependencies evolve:

| Step | Purpose | Command |
|---|---|---|
| Refresh help snapshots | Sync CLI defaults with active binary | `./generate_mmseqs_docs.sh /path/to/mmseqs` |
| Rebuild dependency artifacts | Refresh command topology from source and workflow scripts | `mmseqs2_docs/scripts/build_dependency_graph` |

| Step | Purpose | Command |
|---|---|---|
| Regenerate command reference | Rebuild per-command pages and reference index | `mmseqs2_docs/scripts/generate_command_refer` |
| Regenerate module pages | Rebuild `submodules/*.md` with dependency crosslinks | `mmseqs2_docs/scripts/generate_module_docs.py` |
| Validate structure | Detect missing pages, duplicate command sections, broken links | `mmseqs2_docs/scripts/validate_docs.py` |
| Build deliverable PDF | Produce final manual | `./mmseqs2_docs/build_pdf.sh` |

### 18.6 6. Cross References

Use these chapters together:

| Need | Chapter |
|---|---|
| Storage/index/split/parallel mechanics | `foundations.md` |
| Architecture and dependency navigation | `system_map.md`, `reference/dependency_map.md` |
| Task-oriented command selection | `manual.md`, `submodules/*.md` |
| Full command-level option detail | `reference/index.md`, `reference/<command>.md` |

# 19 Appendix A: Wiki Reference

The legacy wiki remains valuable for historical context, long-form examples, and extended narrative discussion. The primary documentation set in this manual is structured for current command layering, dependency tracing, and module-level navigation.

Storage format, indexing, memory-split behavior, and parallel execution concepts are now summarized canonically in `foundations.md`. Use `wiki.md` for longer historical derivations, older operational examples, and extended FAQ-style material.

| Use Case | Preferred Source |
|---|---|
| Canonical summary of speed mechanics and execution model | `foundations.md` |
| Current command relationships and layer-aware navigation | `manual.md`, `submodules/*.md`, `reference/` |
| Operational sharp bits and expert pipeline guidance | `sharp_bits.md`, `expert_manual.md` |

| Use Case | Preferred Source |
|---|---|
| Historical examples and broader background | `wiki.md` |

# 20 Appendix B: Developer Reference

Developer maintenance should start from generated artifacts, then fall back to narrative chapters when additional context is needed.

| Maintenance Need | Primary Artifact |
|---|---|
| Canonical architecture and performance foundations | `system_map.md`, `foundations.md` |
| Command topology and cascade edges | `reference/dependency_map.md` |
| Command catalog and snapshot coverage | `reference/index.md` |
| Dependency extraction logic | `scripts/build_dependency_graph.py` |
| Command page generation | `scripts/generate_command_reference.py` |
| Functional module generation | `scripts/generate_module_docs.py` |
| Structural consistency checks | `scripts/validate_docs.py` |

Legacy developer-oriented prose is preserved in `developer_manual.md`.

# 21 MMseqs2 Command Reference Index

This reference is generated from source metadata and local help snapshots.

| Metric | Value |
|---|---|
| Total visible commands | `128` |
| Commands with help snapshots | `97` |
| Commands missing snapshots | `31` |

> **Warning**
>
> Some visible commands do not have local help snapshots. Use `generate_mmseqs_docs.sh` to refresh before publishing final CLI defaults.

| Command | Snapshot status |
|---|---|
| `addtaxonomy` | missing |
| `aggregatetax` | missing |
| `aggregatetaxweights` | missing |

| Command | Snapshot status |
|---|---|
| `convert2fasta` | missing |
| `convertca3m` | missing |
| `convertkb` | missing |
| `convertprofiledb` | missing |
| `countkmer` | missing |
| `createbintaxmapping` | missing |
| `createbintaxonomy` | missing |
| `createdmptaxonomy` | missing |
| `databases` | missing |
| `diffseqdbs` | missing |
| `expand2profile` | missing |
| `extractdomains` | missing |
| `fwbw` | missing |
| `gff2db` | missing |
| `majoritylca` | missing |
| `maskbygff` | missing |
| `nrtotaxmapping` | missing |
| `pairaln` | missing |
| `pickconsensusrep` | missing |
| `profile2consensus` | missing |
| `profile2neff` | missing |
| `profile2pssm` | missing |
| `profile2repseq` | missing |
| `rmdb` | missing |
| `sequence2profile` | missing |
| `summarizeheaders` | missing |
| `summarizetabs` | missing |
| `transitivealign` | missing |

Primary maps: Dependency map.

## 21.1 Easy Workflows

| Command | Layer | Snapshot |
|---|---|---|
| easy-cluster | workflow | help |
| easy-linclust | workflow | help |
| easy-linsearch | workflow | help |
| easy-rbh | workflow | help |
| easy-search | workflow | help |
| easy-taxonomy | workflow | help |

## 21.2 Search Workflows

| Command | Layer | Snapshot |
|---|---|---|
| linsearch | high_level_api | help |
| map | high_level_api | help |
| rbh | high_level_api | help |
| search | high_level_api | help |

## 21.3 Clustering

| Command | Layer | Snapshot |
|---|---|---|
| clust | mid_level_api | help |
| cluster | high_level_api | help |
| clusterupdate | high_level_api | help |
| clusthash | mid_level_api | help |
| linclust | high_level_api | help |
| mergeclusters | mid_level_api | help |
| pickconsensusrep | mid_level_api | missing-help |

## 21.4 Prefiltering

| Command | Layer | Snapshot |
|---|---|---|
| countkmer | low_level_api | missing-help |
| gappedprefilter | mid_level_api | help |
| kmermatcher | mid_level_api | help |
| kmersearch | mid_level_api | help |
| prefilter | mid_level_api | help |

| Command | Layer | Snapshot |
|---|---|---|
| ungappedprefilter | mid_level_api | help |

## 21.5 Alignment

| Command | Layer | Snapshot |
|---|---|---|
| align | mid_level_api | help |
| alignall | mid_level_api | help |
| alignbykmer | mid_level_api | help |
| expandaln | mid_level_api | help |
| fwbw | mid_level_api | missing-help |
| offsetalignment | low_level_api | help |
| proteinaln2nucl | low_level_api | help |
| rescorediagonal | mid_level_api | help |
| transitivealign | mid_level_api | missing-help |

## 21.6 Profiles

| Command | Layer | Snapshot |
|---|---|---|
| convertca3m | mid_level_api | missing-help |
| convertmsa | low_level_api | help |
| convertprofiledb | low_level_api | missing-help |
| expand2profile | mid_level_api | missing-help |
| msa2profile | low_level_api | help |
| msa2result | low_level_api | help |
| pairaln | low_level_api | missing-help |
| profile2consensus | low_level_api | missing-help |
| profile2neff | low_level_api | missing-help |
| profile2pssm | low_level_api | missing-help |
| profile2repseq | low_level_api | missing-help |
| result2profile | low_level_api | help |
| sequence2profile | low_level_api | missing-help |
| tsv2exprofiledb | mid_level_api | help |

## 21.7 Database

| Command | Layer | Snapshot |
|---|---|---|
| aliasdb | low_level_api | help |
| concatdbs | low_level_api | help |
| cpdb | low_level_api | help |
| createdb | low_level_api | help |
| createindex | low_level_api | help |
| createlinindex | low_level_api | help |
| createsubdb | low_level_api | help |
| databases | low_level_api | missing-help |
| db2tar | low_level_api | help |
| lndb | low_level_api | help |
| mergedbs | low_level_api | help |
| mvdb | low_level_api | help |
| renamedbkeys | low_level_api | help |
| rmdb | low_level_api | missing-help |
| splitdb | low_level_api | help |
| splitsequence | low_level_api | help |
| subtractdbs | low_level_api | help |
| swapdb | low_level_api | help |
| tar2db | low_level_api | help |
| tsv2db | low_level_api | help |

## 21.8 Result Handling

| Command | Layer | Snapshot |
|---|---|---|
| convert2fasta | low_level_api | missing-help |
| convertalis | low_level_api | help |
| createseqfiledb | low_level_api | help |
| createtsv | low_level_api | help |
| extractdomains | low_level_api | missing-help |
| filterresult | low_level_api | help |
| result2dnamsa | low_level_api | help |
| result2flat | low_level_api | help |

| Command | Layer | Snapshot |
|---|---|---|
| result2msa | low_level_api | help |
| result2rbh | low_level_api | help |
| result2repseq | low_level_api | help |
| result2stats | low_level_api | help |
| sortresult | low_level_api | help |
| summarizealis | low_level_api | help |
| summarizeheaders | low_level_api | missing-help |
| summarizeresult | low_level_api | help |
| swapresults | low_level_api | help |

## 21.9 Sequence Manipulation

| Command | Layer | Snapshot |
|---|---|---|
| extractalignedregion | low_level_api | help |
| extractframes | low_level_api | help |
| extractorfs | low_level_api | help |
| masksequence | low_level_api | help |
| orftocontig | low_level_api | help |
| recoverlongestorf | low_level_api | help |
| reverseseq | low_level_api | help |
| translateaa | low_level_api | help |
| translatenucs | low_level_api | help |

## 21.10 Taxonomy

| Command | Layer | Snapshot |
|---|---|---|
| addtaxonomy | low_level_api | missing-help |
| aggregatetax | low_level_api | missing-help |
| aggregatetaxweights | low_level_api | missing-help |
| createbintaxmapping | low_level_api | missing-help |
| createbintaxonomy | low_level_api | missing-help |
| createdmptaxonomy | low_level_api | missing-help |
| createtaxdb | low_level_api | help |
| filtertaxdb | low_level_api | help |

| Command | Layer | Snapshot |
|---|---|---|
| filtertaxseqdb | low_level_api | help |
| lca | low_level_api | help |
| lcaalign | low_level_api | help |
| majoritylca | low_level_api | missing-help |
| nrtotaxmapping | low_level_api | missing-help |
| taxonomy | high_level_api | help |
| taxonomyreport | low_level_api | help |

## 21.11 Multi Hit

| Command | Layer | Snapshot |
|---|---|---|
| besthitperset | high_level_api | help |
| combinepvalperset | high_level_api | help |
| mergeresultsbyset | high_level_api | help |
| multihitdb | high_level_api | help |
| multihitsearch | high_level_api | help |

## 21.12 Utilities

| Command | Layer | Snapshot |
|---|---|---|
| apply | low_level_api | help |
| compress | low_level_api | help |
| convertkb | low_level_api | missing-help |
| decompress | low_level_api | help |
| diffseqdbs | low_level_api | missing-help |
| filterdb | low_level_api | help |
| gff2db | low_level_api | missing-help |
| gpuserver | low_level_api | help |
| maskbygff | low_level_api | missing-help |
| prefixid | low_level_api | help |
| setextendeddbtype | low_level_api | help |
| suffixid | low_level_api | help |
| summarizetabs | low_level_api | missing-help |
| touchdb | low_level_api | help |

| Command | Layer | Snapshot |
|---|---|---|
| `unpackdb` | `low_level_api` | `help` |
| `view` | `low_level_api` | `help` |

# 22 MMseqs2 Dependency Map

This file is generated from `MMseqs2/src/MMseqsBase.cpp` and workflow scripts.

| Metric | Value |
|---|---|
| Total visible commands | `128` |

`n/a` in connection fields means no direct edge was resolved by static extraction.

## 22.1 Easy Workflows

### 22.1.1 `easy-cluster`

Slower, sensitive clustering.

| Aspect | Value |
|---|---|
| Layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Calls | `cluster`, `createdb`, `createseqfiledb`, `createtsv`, `result2flat`, `result2repseq`, `rmdb` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.1.2 `easy-linclust`

Fast linear time cluster, less sensitive clustering.

| Aspect | Value |
|---|---|
| Layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Calls | `createdb`, `createseqfiledb`, `createtsv`, `linclust`, `result2flat`, `result2repseq`, `rmdb` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |

| Aspect | Value |
|---|---|
| Command reference | Open page |

#### 22.1.3 `easy-linsearch`

Fast, less sensitive homology search.

| Aspect | Value |
|---|---|
| Layer | `workflow` |
| Category flags | `COMMAND_EASY \| COMMAND_EXPERT` |
| Calls | `convertalis`, `createdb`, `createlinindex`, `linsearch`, `rmdb`, `search`, `summarizeresult` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.1.4 `easy-rbh`

Find reciprocal best hit.

| Aspect | Value |
|---|---|
| Layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Calls | `convertalis`, `createdb`, `rbh`, `rmdb` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.1.5 `easy-search`

Sensitive homology search.

| Aspect | Value |
|---|---|
| Layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Calls | `convertalis`, `createdb`, `createlinindex`, `linsearch`, `rmdb`, `search`, `summarizeresult` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |

| Aspect | Value |
|---|---|
| Command reference | Open page |

#### 22.1.6 `easy-taxonomy`

Taxonomic classification.

| Aspect | Value |
|---|---|
| Layer | `workflow` |
| Category flags | `COMMAND_EASY` |
| Calls | `addtaxonomy`, `convertalis`, `createdb`, `createtsv`, `filterdb`, `lca`, `rmdb`, `summarizealis`, `swapresults`, `taxonomy`, `taxonomyreport` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.2 Search Workflows

#### 22.2.1 `linsearch`

Fast, less sensitive homology search.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MAIN\|COMMAND_EXPERT` |
| Calls | `align`, `concatdbs`, `extractorfs`, `filterdb`, `kmersearch`, `offsetalignment`, `rescorediagonal`, `rmdb`, `swapresults` |
| Called by | `easy-linsearch`, `easy-search` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.2.2 `map`

Map nearly identical sequences.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_MAIN` |
| Calls | `search` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.2.3 `rbh`

Reciprocal best hit search.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Calls | `filterdb`, `mergedbs`, `result2rbh`, `rmdb`, `search`, `swapresults` |
| Called by | `easy-rbh` |
| Workflow scripts | `easyrbh.sh` |
| Command reference | Open page |

### 22.2.4 `search`

Sensitive homology search.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Calls | `align`, `createsubdb`, `expand2profile`, `expandaln`, `extractframes`, `extractorfs`, `filterresult`, `lcaalign`, `mergedbs`, `mvdb`, `offsetalignment`, `prefilter`, `profile2consensus`, `rescorediagonal`, `result2profile`, `result2stats`, `rmdb`, `search`, `splitsequence`, `subtractdbs`, `swapresults`, `ungappedprefilter` |
| Called by | `clusterupdate`, `easy-linsearch`, `easy-search`, `map`, `multihitsearch`, `rbh`, `search`, `taxonomy` |

| Aspect | Value |
|---|---|
| Workflow scripts | enrich.sh, iterativepp.sh, map.sh, multihitsearch.sh, rbh.sh, taxonomy.sh, update_clustering.sh |
| Command reference | Open page |

## 22.3 Clustering

### 22.3.1 clust

Cluster result by Set-Cover/Connected-Component/Greedy-Incremental.

| Aspect | Value |
|---|---|
| Layer | mid_level_api |
| Category flags | COMMAND_CLUSTER |
| Calls | n/a |
| Called by | cluster, linclust |
| Workflow scripts | cascaded_clustering.sh, clustering.sh, linclust.sh, nucleotide_clustering.sh |
| Command reference | Open page |

### 22.3.2 cluster

Slower, sensitive clustering.

| Aspect | Value |
|---|---|
| Layer | high_level_api |
| Category flags | COMMAND_MAIN |
| Calls | align, clust, clusthash, concatdbs, createsubdb, extractframes, filterdb, linclust, mergeclusters, mergedbs, mvdb, offsetalignment, prefilter, rescorediagonal, rmdb, subtractdbs, swapdb, tsv2db |
| Called by | clusterupdate, easy-cluster |
| Workflow scripts | update_clustering.sh |
| Command reference | Open page |

### 22.3.3 clusterupdate

Update previous clustering with new sequences.

| Aspect | Value |
|---|---|
| Layer | high_level_api |
| Category flags | COMMAND_MAIN |
| Calls | cluster, concatdbs, createsubdb, diffseqdbs, filterdb, mergedbs, mvdb, prefixid, renamedbkeys, result2repseq, rmdb, search, swapdb |
| Called by | n/a |
| Workflow scripts | n/a |
| Command reference | Open page |

#### 22.3.4 **clusthash**

Hash-based clustering of equal length sequences.

| Aspect | Value |
|---|---|
| Layer | mid_level_api |
| Category flags | COMMAND_CLUSTER |
| Calls | n/a |
| Called by | cluster |
| Workflow scripts | clustering.sh |
| Command reference | Open page |

#### 22.3.5 **linclust**

Fast, less sensitive clustering.

| Aspect | Value |
|---|---|
| Layer | high_level_api |
| Category flags | COMMAND_MAIN |
| Calls | align, clust, createsubdb, filterdb, kmermatcher, mergeclusters, rescorediagonal, rmdb |
| Called by | cluster, easy-linclust |
| Workflow scripts | cascaded_clustering.sh, nucleotide_clustering.sh |
| Command reference | Open page |

#### 22.3.6 `mergeclusters`

Merge multiple cascaded clustering steps.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_CLUSTER` |
| Calls | `n/a` |
| Called by | `cluster`, `linclust` |
| Workflow scripts | `cascaded_clustering.sh`, `clustering.sh`, `linclust.sh`, `nucleotide_clustering.sh` |
| Command reference | Open page |

#### 22.3.7 `pickconsensusrep`

Select new representatives for each cluster based on consensus.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_CLUSTER` |
| Calls | `align`, `msa2profile`, `prefixid`, `renamedbkeys`, `result2msa`, `rmdb`, `tsv2db` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.4 Prefiltering

#### 22.4.1 `countkmer`

Count k-mers.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.4.2 `gappedprefilter`

Optimal Smith-Waterman-based prefiltering (slow).

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.4.3 `kmermatcher`

Find bottom-m-hashed k-mer matches within sequence DB.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Calls | `n/a` |
| Called by | `linclust` |
| Workflow scripts | `linclust.sh` |
| Command reference | Open page |

### 22.4.4 `kmersearch`

Find bottom-m-hashed k-mer matches between target and query DB.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Calls | `n/a` |
| Called by | `linsearch` |
| Workflow scripts | `linsearch.sh` |
| Command reference | Open page |

### 22.4.5 `prefilter`

Double consecutive diagonal k-mer search.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Calls | `n/a` |
| Called by | `cluster`, `search`, `taxonomy` |
| Workflow scripts | `blastp.sh`, `blastpgp.sh`, `cascaded_clustering.sh`, `clustering.sh`, `enrich.sh`, `iterativepp.sh`, `nucleotide_clustering.sh`, `searchslicedtargetprofile.sh`, `searchtargetprofile.sh`, `taxpercontig.sh`, `translated_search.sh` |
| Command reference | Open page |

### 22.4.6 `ungappedprefilter`

Optimal diagonal score search.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_PREFILTER` |
| Calls | `n/a` |
| Called by | `search` |
| Workflow scripts | `blastp.sh`, `blastpgp.sh` |
| Command reference | Open page |

## 22.5 Alignment

### 22.5.1 `align`

Optimal gapped local alignment.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Calls | `n/a` |
| Called by | `cluster`, `linclust`, `linsearch`, `pickconsensusrep`, `search` |

| Aspect | Value |
|---|---|
| Workflow scripts | `iterativepp.sh`, `nucleotide_clustering.sh`, `pickconsensusrep.sh`, `searchslicedtargetprofile.sh` |
| Command reference | Open page |

### 22.5.2 `alignall`

Within-result all-vs-all gapped local alignment.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.5.3 `alignbykmer`

Heuristic gapped local k-mer based alignment.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.5.4 `expandaln`

Expand an alignment result based on another.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_PROFILE_PROFILE` |
| Calls | `n/a` |

| Aspect | Value |
|---|---|
| Called by | `search` |
| Workflow scripts | `enrich.sh`, `iterativepp.sh` |
| Command reference | Open page |

### 22.5.5 `fwbw`

Forward Backward Alignment.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.5.6 `offsetalignment`

Offset alignment by ORF start position.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `cluster`, `linsearch`, `search` |
| Workflow scripts | `blastn.sh`, `linsearch.sh`, `nucleotide_clustering.sh`, `translated_search.sh` |
| Command reference | Open page |

### 22.5.7 `proteinaln2nucl`

Transform protein alignments to nucleotide alignments.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |

| Aspect | Value |
| --- | --- |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.5.8 `rescorediagonal`

Compute sequence identity for diagonal.

| Aspect | Value |
| --- | --- |
| Layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Calls | `n/a` |
| Called by | `cluster`, `linclust`, `linsearch`, `search`, `taxonomy` |
| Workflow scripts | `linclust.sh`, `linsearch.sh`, `nucleotide_clustering.sh`, `taxpercontig.sh` |
| Command reference | Open page |

#### 22.5.9 `transitivealign`

Transfer alignments via transitivity.

| Aspect | Value |
| --- | --- |
| Layer | `mid_level_api` |
| Category flags | `COMMAND_ALIGNMENT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.6 Profiles

#### 22.6.1 `convertca3m`

Convert a cA3M DB to a result DB.

| Aspect | Value |
| --- | --- |
| Layer | `mid_level_api` |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_PROFILE_PROFILE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.6.2 `convertmsa`

Convert Stockholm/PFAM MSA file to a MSA DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION` |
| Calls | `n/a` |
| Called by | `databases` |
| Workflow scripts | `databases.sh` |
| Command reference | Open page |

#### 22.6.3 `convertprofiledb`

Convert a HH-suite HHM DB to a profile DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.6.4 `expand2profile`

Expand an alignment result based on another and create a profile.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_PROFILE_PROFILE` |

| Aspect | Value |
|---|---|
| Calls | n/a |
| Called by | search |
| Workflow scripts | iterativepp.sh |
| Command reference | Open page |

### 22.6.5 msa2profile

Convert a MSA DB to a profile DB.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_PROFILE \| COMMAND_DATABASE_CREATION |
| Calls | n/a |
| Called by | databases, pickconsensusrep |
| Workflow scripts | databases.sh, pickconsensusrep.sh |
| Command reference | Open page |

### 22.6.6 msa2result

Convert a MSA DB to a profile DB.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_PROFILE \| COMMAND_EXPERT |
| Calls | n/a |
| Called by | n/a |
| Workflow scripts | n/a |
| Command reference | Open page |

### 22.6.7 pairaln

Pair sequences to match best protein A and B from a species.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_EXPERT |
| Calls | n/a |

| Aspect | Value |
|---|---|
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.6.8 `profile2consensus`

Extract consensus sequence DB from a profile DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Calls | `n/a` |
| Called by | `search` |
| Workflow scripts | `iterativepp.sh` |
| Command reference | Open page |

#### 22.6.9 `profile2neff`

Convert a profile DB to a tab-separated list of Neff scores.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.6.10 `profile2pssm`

Convert a profile DB to a tab-separated PSSM file.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Calls | `n/a` |
| Called by | `n/a` |

| Aspect | Value |
|---|---|
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.6.11 `profile2repseq`

Extract representative sequence DB from a profile DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.6.12 `result2profile`

Compute profile DB from a result DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Calls | `n/a` |
| Called by | `search` |
| Workflow scripts | `blastpgp.sh`, `enrich.sh` |
| Command reference | Open page |

#### 22.6.13 `sequence2profile`

Turn sequence into profile by adding context specific pseudo counts.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_PROFILE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |

| Aspect | Value |
|---|---|
| Command reference | Open page |

#### 22.6.14 `tsv2exprofiledb`

Create a expandable profile db from TSV files.

| Aspect | Value |
|---|---|
| Layer | `mid_level_api` |
| Category flags | `COMMAND_PROFILE_PROFILE` |
| Calls | `aliasdb`, `compress`, `mvdb`, `rmdb`, `tsv2db` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

## 22.7 Database

#### 22.7.1 `aliasdb`

Create relative symlink of DB to another name in the same folder.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Calls | `n/a` |
| Called by | `tsv2exprofiledb` |
| Workflow scripts | `tsv2exprofiledb.sh` |
| Command reference | Open page |

#### 22.7.2 `concatdbs`

Concatenate two DBs, giving new IDs to entries from 2nd DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SET` |
| Calls | `n/a` |
| Called by | `cluster`, `clusterupdate`, `linsearch` |

| Aspect | Value |
| --- | --- |
| Workflow scripts | linsearch.sh, nucleotide_clustering.sh, update_clustering.sh |
| Command reference | Open page |

### 22.7.3 cpdb

Copy a DB.

| Aspect | Value |
| --- | --- |
| Layer | low_level_api |
| Category flags | COMMAND_STORAGE |
| Calls | n/a |
| Called by | n/a |
| Workflow scripts | n/a |
| Command reference | Open page |

### 22.7.4 createdb

Convert FASTA/Q file(s) to a sequence DB.

| Aspect | Value |
| --- | --- |
| Layer | low_level_api |
| Category flags | COMMAND_DATABASE_CREATION |
| Calls | n/a |
| Called by | databases, easy-cluster, easy-linclust, easy-linsearch, easy-rbh, easy-search, easy-taxonomy, multihitdb |
| Workflow scripts | databases.sh, easycluster.sh, easyrbh.sh, easysearch.sh, easytaxonomy.sh, multihitdb.sh |
| Command reference | Open page |

### 22.7.5 createindex

Store precomputed index on disk to reduce search overhead.

| Aspect | Value |
| --- | --- |
| Layer | low_level_api |
| Category flags | COMMAND_DATABASE_CREATION |

| Aspect | Value |
|---|---|
| Calls | extractframes, extractorfs, rmdb, splitsequence |
| Called by | n/a |
| Workflow scripts | n/a |
| Command reference | Open page |

#### 22.7.6 createlinindex

Create linsearch index.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_DATABASE_CREATION \| COMMAND_EXPERT |
| Calls | extractframes, extractorfs, rmdb, splitsequence |
| Called by | easy-linsearch, easy-search |
| Workflow scripts | easysearch.sh |
| Command reference | Open page |

#### 22.7.7 createsubdb

Create a subset of a DB from list of DB keys.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_SET |
| Calls | n/a |
| Called by | cluster, clusterupdate, linclust, search, taxonomy |
| Workflow scripts | blastp.sh, cascaded_clustering.sh, clustering.sh, linclust.sh, nucleotide_clustering.sh, taxpercontig.sh, translated_search.sh, update_clustering.sh |
| Command reference | Open page |

#### 22.7.8 `databases`

List and download databases.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION` |
| Calls | `convertmsa`, `createdb`, `createtaxdb`, `msa2profile`, `nrtotaxmapping`, `prefixid`, `rmdb`, `tar2db` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.7.9 `db2tar`

Archive contents of a DB to a tar archive.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DATABASE_CREATION \| COMMAND_EXPERT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.7.10 `lndb`

Symlink a DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.7.11 `mergedbs`

Merge entries from multiple DBs.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SET` |
| Calls | `n/a` |
| Called by | `cluster`, `clusterupdate`, `rbh`, `search` |
| Workflow scripts | `blastp.sh`, `blastpgp.sh`, `cascaded_clustering.sh`, `enrich.sh`, `iterativepp.sh`, `rbh.sh`, `searchslicedtargetprofile.sh`, `update_clustering.sh` |
| Command reference | Open page |

#### 22.7.12 `mvdb`

Move a DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Calls | `n/a` |
| Called by | `cluster`, `clusterupdate`, `search`, `taxonomy`, `tsv2exprofiledb` |
| Workflow scripts | `blastp.sh`, `cascaded_clustering.sh`, `searchslicedtargetprofile.sh`, `taxonomy.sh`, `tsv2exprofiledb.sh`, `update_clustering.sh` |
| Command reference | Open page |

#### 22.7.13 `renamedbkeys`

Create a new DB with original keys renamed.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Calls | `n/a` |
| Called by | `clusterupdate`, `pickconsensusrep` |
| Workflow scripts | `update_clustering.sh` |

| Aspect | Value |
|---|---|
| Command reference | Open page |

#### 22.7.14 `rmdb`

Remove a DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Calls | `n/a` |
| Called by | `cluster`, `clusterupdate`, `createindex`, `createlinindex`, `databases`, `easy-cluster`, `easy-linclust`, `easy-linsearch`, `easy-rbh`, `easy-search`, `easy-taxonomy`, `linclust`, `linsearch`, `multihitsearch`, `pickconsensusrep`, `rbh`, `search`, `taxonomy`, `tsv2exprofiledb` |
| Workflow scripts | `blastn.sh`, `blastp.sh`, `blastpgp.sh`, `cascaded_clustering.sh`, `clustering.sh`, `createindex.sh`, `databases.sh`, `easycluster.sh`, `easyrbh.sh`, `easysearch.sh`, `easytaxonomy.sh`, `iterativepp.sh`, `linclust.sh`, `linsearch.sh`, `multihitsearch.sh`, `nucleotide_clustering.sh`, `pickconsensusrep.sh`, `rbh.sh`, `searchslicedtargetprofile.sh`, `searchtargetprofile.sh`, `taxonomy.sh`, `taxpercontig.sh`, `translated_search.sh`, `tsv2exprofiledb.sh`, `update_clustering.sh` |
| Command reference | Open page |

#### 22.7.15 `splitdb`

Split DB into subsets.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SET` |

| Aspect | Value |
|---|---|
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.7.16 `splitsequence`

Split sequences by length.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Calls | `n/a` |
| Called by | `createindex`, `createlinindex`, `search` |
| Workflow scripts | `blastn.sh`, `createindex.sh` |
| Command reference | Open page |

#### 22.7.17 `subtractdbs`

Remove all entries from first DB occurring in second DB by key.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SET` |
| Calls | `n/a` |
| Called by | `cluster`, `search` |
| Workflow scripts | `blastpgp.sh`, `cascaded_clustering.sh`, `enrich.sh`, `iterativepp.sh`, `nucleotide_clustering.sh` |
| Command reference | Open page |

#### 22.7.18 `swapdb`

Transpose DB with integer values in first column.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DB` |

| Aspect | Value |
|---|---|
| Calls | n/a |
| Called by | cluster, clusterupdate, multihitdb, taxonomy |
| Workflow scripts | cascaded_clustering.sh, multihitdb.sh, taxpercontig.sh, update_clustering.sh |
| Command reference | Open page |

#### 22.7.19 tar2db

Convert content of tar archives to any DB.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_DATABASE_CREATION \| COMMAND_EXPERT |
| Calls | n/a |
| Called by | databases |
| Workflow scripts | databases.sh |
| Command reference | Open page |

#### 22.7.20 tsv2db

Convert a TSV file to any DB.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_DATABASE_CREATION \| COMMAND_EXPERT |
| Calls | n/a |
| Called by | cluster, multihitdb, pickconsensusrep, tsv2exprofiledb |
| Workflow scripts | cascaded_clustering.sh, multihitdb.sh, pickconsensusrep.sh, tsv2exprofiledb.sh |
| Command reference | Open page |

### 22.8 Result Handling

#### 22.8.1 convert2fasta

Convert sequence DB to FASTA format.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.8.2 `convertalis`

Convert alignment DB to BLAST-tab, SAM or custom format.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION` |
| Calls | `n/a` |
| Called by | `easy-linsearch`, `easy-rbh`, `easy-search`, `easy-taxonomy` |
| Workflow scripts | `easyrbh.sh`, `easysearch.sh`, `easytaxonomy.sh` |
| Command reference | Open page |

### 22.8.3 `createseqfiledb`

Create a DB of unaligned FASTA entries.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION \| COMMAND_EXPERT` |
| Calls | `n/a` |
| Called by | `easy-cluster`, `easy-linclust` |
| Workflow scripts | `easycluster.sh` |
| Command reference | Open page |

### 22.8.4 `createtsv`

Convert result DB to tab-separated flat file.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION` |
| Calls | `n/a` |
| Called by | `easy-cluster`, `easy-linclust`, `easy-taxonomy` |
| Workflow scripts | `easycluster.sh`, `easytaxonomy.sh` |
| Command reference | Open page |

### 22.8.5 `extractdomains`

Extract highest scoring alignment regions for each sequence from BLAST-tab file.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.8.6 `filterresult`

Pairwise alignment result filter.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `search` |
| Workflow scripts | `searchslicedtargetprofile.sh` |
| Command reference | Open page |

### 22.8.7 `result2dnamsa`

Compute MSA DB with out insertions in the query for DNA sequences.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |

| Aspect | Value |
| --- | --- |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.8.8 `result2flat`

Create flat file by adding FASTA headers to DB entries.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_FORMAT_CONVERSION \| COMMAND_EXPERT` |
| Calls | `n/a` |
| Called by | `easy-cluster`, `easy-linclust` |
| Workflow scripts | `easycluster.sh` |
| Command reference | Open page |

#### 22.8.9 `result2msa`

Compute MSA DB from a result DB.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `pickconsensusrep` |
| Workflow scripts | `pickconsensusrep.sh` |
| Command reference | Open page |

#### 22.8.10 `result2rbh`

Filter a merged result DB to retain only reciprocal best hits.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |

| Aspect | Value |
|---|---|
| Calls | `n/a` |
| Called by | `rbh` |
| Workflow scripts | `rbh.sh` |
| Command reference | Open page |

#### 22.8.11 `result2repseq`

Get representative sequences from result DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `clusterupdate`, `easy-cluster`, `easy-linclust` |
| Workflow scripts | `easycluster.sh`, `update_clustering.sh` |
| Command reference | Open page |

#### 22.8.12 `result2stats`

Compute statistics for each entry in a DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `multihitdb`, `search` |
| Workflow scripts | `multihitdb.sh`, `searchslicedtargetprofile.sh` |
| Command reference | Open page |

#### 22.8.13 `sortresult`

Sort a result DB in the same order as the prefilter or align module.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |

| Aspect | Value |
| --- | --- |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.8.14 `summarizealis`

Summarize alignment result to one row (uniq. cov., cov., avg. seq. id.).

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `easy-taxonomy` |
| Workflow scripts | `easytaxonomy.sh` |
| Command reference | Open page |

### 22.8.15 `summarizeheaders`

Summarize FASTA headers of result DB.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.8.16 `summarizeresult`

Extract annotations from alignment DB.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `easy-linsearch`, `easy-search` |

| Aspect | Value |
|---|---|
| Workflow scripts | `easysearch.sh` |
| Command reference | Open page |

#### 22.8.17 **`swapresults`**

Transpose prefilter/alignment DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_RESULT` |
| Calls | `n/a` |
| Called by | `easy-taxonomy`, `linsearch`, `rbh`, `search` |
| Workflow scripts | `easytaxonomy.sh`, `linsearch.sh`, `rbh.sh`, `searchslicedtargetprofile.sh`, `searchtargetprofile.sh` |
| Command reference | Open page |

### 22.9 Sequence Manipulation

#### 22.9.1 **`extractalignedregion`**

Extract aligned sequence region from query.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.9.2 **`extractframes`**

Extract frames from a nucleotide sequence DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Calls | `n/a` |

| Aspect | Value |
|---|---|
| Called by | `cluster`, `createindex`, `createlinindex`, `search` |
| Workflow scripts | `blastn.sh`, `createindex.sh`, `nucleotide_clustering.sh`, `translated_search.sh` |
| Command reference | Open page |

#### 22.9.3 `extractorfs`

Six-frame extraction of open reading frames.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Calls | `n/a` |
| Called by | `createindex`, `createlinindex`, `linsearch`, `multihitdb`, `search`, `taxonomy` |
| Workflow scripts | `createindex.sh`, `multihitdb.sh`, `taxpercontig.sh`, `translated_search.sh` |
| Command reference | Open page |

#### 22.9.4 `masksequence`

Soft mask sequence DB using tantan.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.9.5 `orftocontig`

Write ORF locations in alignment format.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_SEQUENCE` |
| Calls | `n/a` |
| Called by | `multihitdb` |
| Workflow scripts | `multihitdb.sh` |
| Command reference | Open page |

#### 22.9.6 `recoverlongestorf`

Recover longest ORF for taxonomy annotation after elimination.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_EXPERT` |
| Calls | `n/a` |
| Called by | `taxonomy` |
| Workflow scripts | `taxpercontig.sh` |
| Command reference | Open page |

#### 22.9.7 `reverseseq`

Reverse (without complement) sequences.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.9.8 `translateaa`

Translate proteins to lexicographically lowest codons.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |

| Aspect | Value |
|---|---|
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.9.9 `translatenucs`

Translate nucleotides to proteins.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SEQUENCE` |
| Calls | `n/a` |
| Called by | `multihitdb` |
| Workflow scripts | `multihitdb.sh` |
| Command reference | Open page |

### 22.10 Taxonomy

#### 22.10.1 `addtaxonomy`

Add taxonomic labels to result DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Calls | `n/a` |
| Called by | `easy-taxonomy` |
| Workflow scripts | `easytaxonomy.sh` |
| Command reference | Open page |

#### 22.10.2 `aggregatetax`

Aggregate multiple taxon labels to a single label.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |

| Aspect | Value |
| --- | --- |
| Calls | `n/a` |
| Called by | `taxonomy` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.10.3 `aggregatetaxweights`

Aggregate multiple taxon labels to a single label.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Calls | `n/a` |
| Called by | `taxonomy` |
| Workflow scripts | `taxpercontig.sh` |
| Command reference | Open page |

#### 22.10.4 `createbintaxmapping`

Create binary taxonomy mapping from tabular taxonomy mapping.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.10.5 `createbintaxonomy`

Create binary taxonomy from NCBI input.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Calls | `n/a` |

| Aspect | Value |
| --- | --- |
| Called by | `createtaxdb` |
| Workflow scripts | `createtaxdb.sh` |
| Command reference | Open page |

#### 22.10.6 `createdmptaxonomy`

Create dmp files from binary taxonomy.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.10.7 `createtaxdb`

Add taxonomic labels to sequence DB.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Calls | `createbintaxonomy` |
| Called by | `databases` |
| Workflow scripts | `databases.sh` |
| Command reference | Open page |

#### 22.10.8 `filtertaxdb`

Filter taxonomy result database.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Calls | `n/a` |
| Called by | `n/a` |

| Aspect | Value |
| --- | --- |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.10.9 `filtertaxseqdb`

Filter taxonomy sequence database.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.10.10 `lca`

Compute the lowest common ancestor.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Calls | `n/a` |
| Called by | `easy-taxonomy`, `taxonomy` |
| Workflow scripts | `taxonomy.sh` |
| Command reference | Open page |

### 22.10.11 `lcaalign`

Efficient gapped alignment for lca computation.

| Aspect | Value |
| --- | --- |
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY` |
| Calls | `n/a` |
| Called by | `search` |
| Workflow scripts | `n/a` |

| Aspect | Value |
|---|---|
| Command reference | Open page |

### 22.10.12 `majoritylca`

Compute the lowest common ancestor using majority voting.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_EXPERT` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.10.13 `nrtotaxmapping`

Create taxonomy mapping for NR database.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Calls | `n/a` |
| Called by | `databases` |
| Workflow scripts | `databases.sh` |
| Command reference | Open page |

### 22.10.14 `taxonomy`

Taxonomic classification.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MAIN` |
| Calls | `aggregatetax`, `aggregatetaxweights`, `createsubdb`, `extractorfs`, `filterdb`, `lca`, `mergeresultsbyset`, `mvdb`, `prefilter`, `recoverlongestorf`, `rescorediagonal`, `rmdb`, `search`, `swapdb`, `taxonomy` |

| Aspect | Value |
|---|---|
| Called by | `easy-taxonomy`, `taxonomy` |
| Workflow scripts | `easytaxonomy.sh`, `taxpercontig.sh` |
| Command reference | Open page |

### 22.10.15 `taxonomyreport`

Create a taxonomy report in Kraken or Krona format.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_TAXONOMY \| COMMAND_FORMAT_CONVERSION` |
| Calls | `n/a` |
| Called by | `easy-taxonomy` |
| Workflow scripts | `easytaxonomy.sh` |
| Command reference | Open page |

## 22.11 Multi Hit

### 22.11.1 `besthitperset`

For each set of sequences compute the best element and update p-value.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Calls | `n/a` |
| Called by | `multihitsearch` |
| Workflow scripts | `multihitsearch.sh` |
| Command reference | Open page |

### 22.11.2 `combinepvalperset`

For each set compute the combined p-value.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Calls | `n/a` |

| Aspect | Value |
|---|---|
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.11.3 `mergeresultsbyset`

Merge results from multiple ORFs back to their respective contig.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Calls | `n/a` |
| Called by | `multihitsearch`, `taxonomy` |
| Workflow scripts | `multihitsearch.sh`, `taxpercontig.sh` |
| Command reference | Open page |

#### 22.11.4 `multihitdb`

Create sequence DB for multi hit searches.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |
| Calls | `createdb`, `extractorfs`, `filterdb`, `orftocontig`, `result2stats`, `swapdb`, `translatenucs`, `tsv2db` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.11.5 `multihitsearch`

Search with a grouped set of sequences against another grouped set.

| Aspect | Value |
|---|---|
| Layer | `high_level_api` |
| Category flags | `COMMAND_MULTIHIT` |

| Aspect | Value |
|---|---|
| Calls | `besthitperset`, `mergeresultsbyset`, `rmdb`, `search` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

## 22.12 Utilities

### 22.12.1 `apply`

Execute given program on each DB entry.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.12.2 `compress`

Compress DB entries.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Calls | `n/a` |
| Called by | `tsv2exprofiledb` |
| Workflow scripts | `tsv2exprofiledb.sh` |
| Command reference | Open page |

### 22.12.3 `convertkb`

Convert UniProtKB data to a DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_SPECIAL` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.12.4 **`decompress`**

Decompress DB entries.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.12.5 **`diffseqdbs`**

Compute diff of two sequence DBs.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Calls | `n/a` |
| Called by | `clusterupdate` |
| Workflow scripts | `update_clustering.sh` |
| Command reference | Open page |

#### 22.12.6 **`filterdb`**

DB filtering by given conditions.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DB` |

| Aspect | Value |
|---|---|
| Calls | `n/a` |
| Called by | `cluster`, `clusterupdate`, `easy-taxonomy`, `linclust`, `linsearch`, `multihitdb`, `rbh`, `taxonomy` |
| Workflow scripts | `cascaded_clustering.sh`, `easytaxonomy.sh`, `linclust.sh`, `linsearch.sh`, `multihitdb.sh`, `rbh.sh`, `taxonomy.sh`, `taxpercontig.sh`, `update_clustering.sh` |
| Command reference | Open page |

#### 22.12.7 `gff2db`

Extract regions from a sequence database based on a GFF3 file.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.12.8 `gpuserver`

Start a GPU server.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_STORAGE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.12.9 `maskbygff`

Mask out sequence regions in a sequence DB by features selected from a GFF3 file.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_SPECIAL` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.12.10 `prefixid`

For each entry in a DB prepend the entry key to the entry itself.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Calls | `n/a` |
| Called by | `clusterupdate`, `databases`, `pickconsensusrep` |
| Workflow scripts | `databases.sh`, `pickconsensusrep.sh`, `update_clustering.sh` |
| Command reference | Open page |

#### 22.12.11 `setextendeddbtype`

Write an extended DB.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

#### 22.12.12 `suffixid`

For each entry in a DB append the entry key to the entry itself.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_DB |
| Calls | n/a |
| Called by | n/a |
| Workflow scripts | n/a |
| Command reference | Open page |

#### 22.12.13 **summarizetabs**

Extract annotations from HHblits BLAST-tab-formatted results.

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_SPECIAL |
| Calls | n/a |
| Called by | n/a |
| Workflow scripts | n/a |
| Command reference | Open page |

#### 22.12.14 **touchdb**

Preload DB into memory (page cache).

| Aspect | Value |
|---|---|
| Layer | low_level_api |
| Category flags | COMMAND_STORAGE |
| Calls | n/a |
| Called by | n/a |
| Workflow scripts | n/a |
| Command reference | Open page |

#### 22.12.15 **unpackdb**

Unpack a DB into separate files.

| Aspect | Value |
|---|---|
| Layer | low_level_api |

| Aspect | Value |
|---|---|
| Category flags | `COMMAND_STORAGE` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |

### 22.12.16 `view`

Print DB entries given in –id-list to stdout.

| Aspect | Value |
|---|---|
| Layer | `low_level_api` |
| Category flags | `COMMAND_DB` |
| Calls | `n/a` |
| Called by | `n/a` |
| Workflow scripts | `n/a` |
| Command reference | Open page |